



Decennial Transformation and Application Modernization (DTAM)

Draft Order PWS
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1.0 Background

The Decennial Census Programs Directorate conducts a census of population and housing, surveys America's communities, and provides enabling geospatial data and data dissemination support to provide quality statistical information to the President and the American people about our changing society.

The Decennial Information Technology (IT) Strategy is centered on the mission to "Provide innovative and secure technologies that enable a cost-effective and efficient 2030 Census for the dissemination of high-quality data to all stakeholders." Accordingly, its vision is "To create an innovative, cost-effective solution that maximizes the value of IT to modernize the science of census taking."

To achieve this mission, the Decennial Census Programs must manage the development and integration of a diverse portfolio of IT solutions, known as the System of Systems (SoS), that includes internally purpose-build Decennial systems, Census Enterprise Systems, and third party contracted systems and services.

Decennial Census Programs use the following guiding principles:

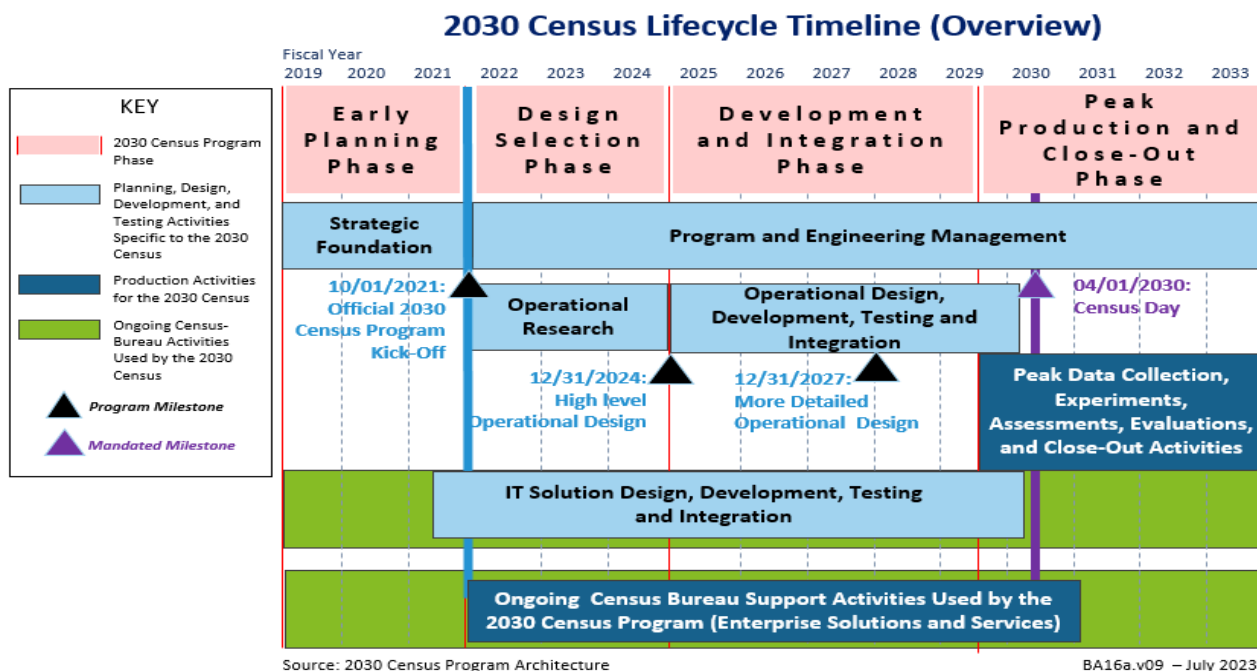
- ***Use the Right People at the Right Time in the Right Positions*** - This is inclusive of the use of federal and contractor staff to meet the staffing needs for architecture, engineering, data science, integration, and delivery so the right technical, management and interpersonal skills are employed for maximized solution delivery.
- ***Collaborate with Critical Stakeholders*** - Decennial operations require close coordination with the Decennial Census Management Division, the Decennial Contracts Execution Office, and other partners to reduce implementation risk across the 2030 Census lifecycle. Integrated cross-functional teams will work together early to refine requirements, inform design decisions, and streamline development to maximize benefits while minimizing duplication.
- ***Integration and Maturation*** – The Census Bureau will advance through iterative development and lessons learned from field tests, allowing emerging technologies to be integrated and solutions to mature as requirements evolve.
- ***Holistic Approach to Design*** - Individual solutions must be developed with overall SoS architecture in mind so systems support each other, are cost-effective, and scalable to meet peak and non-peak performance needs.
- ***Follow Disciplined Program and Engineering Management Practices*** - We follow disciplined, data-driven program and engineering management practices aligned with 2030 policies, focusing on risk, change, schedule, contract, cost, solution development, and stakeholder engagement.
- ***Maximize Reuse*** – By maximizing the reuse of artifacts, processes, and systems, Decennial Census Programs has the opportunity to increase efficiencies without introducing new risks.
- ***Expanded Use of Administrative Data and Advanced Data Science*** – The United States Census Bureau (USCB) is a data-rich environment allowing for expansive use of administrative data leading to improved operational efficiencies and data quality. Administrative data refers to data collected and maintained by Federal, State, and local governments, as well as some commercial entities. This guiding principle is an extension

of the USCB's digital transformation effort to create data products with more existing sources of data to improve accuracy and reduce operational costs. This will be accomplished through expanded use of artificial intelligence and advanced data science disciplines.

- **Use of Secure Cloud and Secure Technologies** – By implementing a cloud-smart strategy, Decennial Census Programs will expand upon the use of secure cloud solutions used during the 2020 Census. Decennial Census Programs will expand its use of open-source technologies, cloud native development best practices, and other industry best practices that provide cost-effective, scalable, secure solutions.

1.1 2030 Census Program Life Cycle

Planning for the 2030 Census began in 2019. The 2030 Census Program is being conducted in four phases, beginning with Early Planning (FY19-FY21), followed by Design Selection (FY22-FY24), Development and Integration (FY25-FY29), and Peak Production and Close-Out (FY29-FY33). Operational design development begins in the Design Selection Phase and is critical to the successful execution of a high-quality census of population and housing. In turn, successful execution is dependent on an evidence-based, integrated operational design that addresses the changing environment and inherent logistical challenges of such a large and complex undertaking. The following graphic shows the four phases of the 2030 Census life cycle, and the type of work and key milestones associated with each phase. Future Census cycles will have their own lifecycles and may or may not include solutions used for the 2030 Census.



1.2 American Community Survey

The American Community Survey (ACS) is a foundational component of the Census Bureau's data infrastructure, providing annually updated, detailed demographic, social, economic, and

housing statistics for communities across the United States and Puerto Rico. Conducted continuously since its full implementation in 2005, the ACS replaces the long-form decennial census by collecting data from a sample of approximately 3.5 million housing units and 150,000 group quarters each year. Its design enables the production of 1-year and 5-year estimates, supporting a wide range of federal, state, and local planning efforts. The ACS informs the allocation of trillions of dollars in federal funding and underpins more than 300 known federal uses. Its integration into the broader decennial census program ensures that operational planning and design decisions are grounded in current, granular data, enhancing the precision and responsiveness of census activities in an evolving demographic landscape.

2.0 Scope

The Decennial Transformation and Application Modernization (DTAM) Contractor will support the Decennial Census Program's technology transformation into a data centric, modern, cloud-based IT solutions. The services provided will assist in the modernization of existing applications as well as the development of new applications.

The goals for this call order include:

- Modernizing USCB application landscape to support Decennial Census Programs so that lifecycle costs and technical debt are minimized.
- Optimizing application architectures to take advantage of the benefits of cloud computing for large scale / real time data processing and use of Artificial Intelligence and Machine Learning (AI/ML), ultimately resulting in Census' ability to increase the speed, quality, and quantity of the data analytics used in the planning and execution of Decennial Census Programs.
- Enhancing and optimizing Decennial Census Program data repositories through innovative design, strategic architecture, and construction methodologies, aimed at eliminating duplication, streamlining access, optimizing storage and fortifying security measures.
- Identifying, designing, and/or coordinating opportunities to improve business outcomes leveraged with data-driven technology solutions.
- Transforming to a cloud-smart organization, to include organizational change management and knowledge training / transferring of cloud capabilities and processes to the federal workforce. This focus includes development and/or refactoring of applications to be cloud native.
- Transforming Decennial Census Programs to a DevSecOps centric organization that provides the highest levels of security, quality, and speed for software engineering.
- Transforming Decennial Census Programs to use scaled agile for software engineering.
- Providing a collaborative effort across contractors, Census federal personnel, and other stakeholders.

To meet these goals, the DTAM Contractor will provide IT architects, IT engineers, data scientists, software developers, testers, IT project management and other technical resources to implement the modernization of existing applications and the development of new applications in order to transform the data management and processing ecosystem supporting Decennial Census Programs. All software development and data rights from this contract will fall under the USCB's ownership.

Key focus areas include but are not limited to:

- Standardized architecture allowing for efficient and accurate sending, receiving, and interpretation of data.
- Applications and architectures supporting high volume and velocity data inputs.
- Applications and architectures supporting the delineation and processing of geographic areas to facilitate the tabulation and dissemination of Census data.
- Data ecosystems and applications supporting near-real time processing of data at scale.
- Data quality and data integrity embedded as a fundamental element of design for data inputs and real-time processing applications.
- Applications allowing for early identification and remediation, with analytics and reporting for traceability and coordination with other quality initiatives across the Decennial.
- Use of business rules and analytics to make data driven decisions.
- Standardized IT Program/Project Management, requirements management, systems engineering, and operational delivery support spanning the lifecycle of the systems supporting Decennial Census Programs.

3.0 Task Areas

The Contractor shall develop solutions and/or provide the capabilities and support in the following functional task areas:

- Task Area 1 –Order and Program Management
- Task Area 2 – Project and Program Management
- Task Area 3 – Architecture and Engineering
- Task Area 4 – Application Development and Management Services
- Task Area 5 – Advanced Data Science and Data Management

The requirements listed in Task Area 1 – Task Area 5 will be further expounded on in Technical Directives (TDs).

3.1 Order and Program Management

The Contractor shall provide a single, coordinated management framework that links order administration and program execution oversight so that planning, decisions, controls, and performance measurement are coherent, compliant, efficient, and measurable across the entire effort.

All deliverables and work products under this order shall be developed, produced, maintained, and submitted in full accordance with the requirements, formats, acceptance criteria, and performance standards established in Section F – Deliverables and Performance and the Performance Matrix.

3.1.1 Order Management

The Contractor shall provide order management activities to support the establishment and execution of governance, management processes, administrative controls, and operations necessary to manage and coordinate all order-administrative tasks as described below.

3.1.1.1 Order Governance

The Contractor shall support the establishment and operation of the organizational structure, management processes, controls, and documentation practices required to oversee order execution and ensure compliance with applicable standards, policies, and Government procedures.

The Contractor duties shall include, but are not limited to:

1. Providing the overall management framework, governance, structure, and controls to effectively manage the requirements of this order.
2. Providing a Contractor Program Management Office (CPMO) dedicated exclusively to this order, including a Program Manager and a designated back-up, responsible for executing all management aspects; providing centralized oversight and communication; and ensuring contractor compliance with the requirements.
3. Developing, documenting, implementing, maintaining, and adhering to a Contract Management Plan (CMP).
4. Managing all technical and administrative requirements in alignment with recognized project management standards, including the latest publication of the Project Management Institute's (PMI) *The Guide to the Project Management Body of Knowledge* (PMBOK®), and implementing processes that incorporate governance and decision-making protocols; schedule, cost, and quality controls; issue and risk management; deliverable tracking; change management; and compliance verification.
5. Obtaining formal Contracting Officer (CO) approval before implementing any change to scope, cost, or schedule, and maintaining a detailed Change Log.

3.1.1.2 Staffing and Subcontractor Management

The Contractor shall support the recruiting, onboarding, supervision, retention and oversight of all personnel and subcontractor resources required for performance under this order.

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, and maintaining a Staffing Plan and Subcontractor Management Plan that is to be submitted as a part of the CMP.
2. Developing, delivering, and maintaining a comprehensive Staffing Matrix that is to be submitted as a part of the Monthly Contract Administration Status Report (MCASR).
3. Maintaining an up-to-date Internal Use Software (IUS) report that is to be submitted as a part of the MCASR.
4. Recruiting, vetting, assigning, managing, and supervising all personnel working under this order.
5. Providing only **U.S. Citizens** who are capable of obtaining a favorable USCB suitability assessment.

6. Establishing and executing onboarding processes for all personnel, including required documentation, suitability submissions, and access requirements.
7. Initiating and monitoring the Census Investigative Services (CIS) suitability assessment, including conducting quality assurance reviews of all suitability documentation prior to submission to CIS to ensure accuracy, completeness, and timeliness.
8. Obtaining COR notification of a favorable suitability assessment for each staff member prior to commencing work, accessing Government facilities or systems, or charging labor to the order.
9. Providing notification, in writing, to the Contracting Officer's Representative (COR) of personnel being offboarded as soon as possible but no later than twenty-four (24) hours after the exit.
10. Retaining personnel-related documentation and providing such records to the Government upon request.
11. Maintaining the required level of expertise and competency of all personnel by ensuring individuals remain current with evolving technologies and by assuming full responsibility for role-appropriate training and requiring personnel to maintain required professional certifications, accreditations, and proficiencies, at no additional cost to the Government.
12. Providing oversight and tracking for all mandatory USCB and Decennial Program training, ensuring all personnel complete required training prior to deadlines specified by the COR.
13. Providing the ability to meet surge requirements by rapidly augmenting staff and responding to unforeseen, ad hoc, or shifting priorities.
14. Assisting the Government with the timely tracking and receipt of Government Furnished Equipment/Government Furnished Information (GFE/GFI), Government system accesses, passwords, etc.
15. Complying with USCB physical and personnel security guidelines.

3.1.1.3 Order Risk Management

The Contractor shall support the proactive identification, assessment, mitigation, monitoring, and escalation of risks and issues that could impact contractual performance, cost, schedule, quality, or compliance. Order risks reported under this subsection shall be included in the Monthly Issue Risk & Issue Report.

The Contractor duties shall include, but are not limited to:

1. Maintaining an active, documented risk management process, including identification, assessment, prioritization, mitigation, and monitoring of all order-execution risks.
2. Developing and maintaining a Risk Register.
3. Developing and executing risk mitigation and/or avoidance plans in coordination with the COR and/or Contracting Officer (CO).
4. Reporting risks, issues, and mitigation plans in all required weekly and monthly reports.
5. Providing immediate notification to the COR and/or CO of any urgent or high-impact risks affecting cost, schedule, performance, compliance, or continuity.

3.1.1.4 Communication Management

The Contractor shall support the establishment of clear, timely, and collaborative communication channels, coordination mechanisms, and meeting cadences between the Contractor and Government to support effective order administration, decision-making, coordination, and documentation.

The Contractor duties shall include, but are not limited to:

1. Establishing and maintaining a Communication Plan.
2. Providing timely notifications of all scheduled and unscheduled events or issues impacting performance.
3. Establishing escalation and crisis-handling procedures for root cause analysis and resolution, to the Government's satisfaction, of major issues.
4. Utilizing Government-approved communication tools and file formats for all reporting and information exchange.
5. Maintaining a Communications and Decision Log to track key communications and decisions.

3.1.2 Program Management

The Contractor shall provide program management activities to support integrated planning, monitoring, control, and coordination of program execution across schedule, quality, risk, readiness, and cross-team collaboration as described below:

3.1.2.1 Program Governance

The Contractor shall support the Contractor's use of Government-issued program governance artifacts, standards, and processes to ensure that all program execution activities conform to Decennial requirements, operational frameworks and oversight expectations.

The Contractor duties shall include, but are not limited to:

1. Leveraging and aligning with USCB and Decennial governing artifacts, including the: 2030 Census Integration & Implementation Plan (IIP) (Attachment J-XX); 2030 Census Operational Delivery Management Plan (ODMP) (Attachment J-XX); the 2030 Census Test & Evaluation Master Plan (TEMP) (Attachment J-XX); 2030 Census IT Defect Management Plan (Attachment J-XX); Decennial Systems Engineering Management Plan (SEMP) (Attachment J-XX); 2030 Census Program Requirements Management Plan (REMP) (Attachment J-XX); and all other applicable Decennial program documents, plans, standards and governance requirements.
2. Implementing and executing program change management processes.
3. Storing all deliverables, work products, artifacts, and supporting documentation in the Government-specified document repository accessible to both the CPMO and the Government Program Management Office (GPMO).

3.1.2.2 Schedule Management

The Contractor shall support the development, integration, and maintenance of schedule artifacts and planning inputs required to coordinate work across the order, including the alignment of the Contractor Work Breakdown Structure (CWBS), Decennial Work Breakdown

Structure (DWBS) (Attachment J-XX), and the Decennial Integrated Master Schedule (IMS) (Attachment J-XX).

3.1.2.2.1 Contractor Work Breakdown Structure (CWBS)

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, and maintaining a detailed CWBS.
2. Tracking, managing, and reporting all contractual deadlines, deliverable due dates, and administrative schedule commitments.
3. Providing and maintaining a CWBS Dictionary.

3.1.2.2.2 Decennial Integrated Master Schedule (IMS) Inputs and Schedule Integration

The Contractor duties shall include, but are not limited to:

1. Providing all required schedule inputs to the Government for incorporation into the Decennial IMS, ensuring completeness, accuracy, and traceability to the CWBS and associated deliverables.
2. Assigning unique CWBS identifiers to each schedule activity and milestone to maintain one-to-one traceability between the schedule, CWBS, deliverables, risks, and reporting.

3.1.2.2.3 Contractor Schedule

The Contractor duties shall include, but are not limited to:

1. Developing and maintaining an order-level schedule baseline.
2. Reporting schedule variances and schedule risk mitigation strategies.

3.1.2.3 Program Quality Management & QASP Performance

The Contractor shall support the monitoring, control, and continuous improvement of program execution and compliance with Government quality surveillance requirements, including development and maintenance of the Quality Control Plan (QCP), participation in Quality Assurance Surveillance Plan (QASP) (Attachment J-XX) activities and program-level defect prevention and resolution.

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, maintaining, and adhering to a QCP.
2. Assessing and documenting quality assurance/quality control lessons learned throughout the lifecycle of the order.
3. Identifying and reporting risks or issues that may impact QASP performance metrics, including staffing risks, workload changes, system constraints, and process challenges, along with mitigation strategies.
4. Providing advanced notice of any anticipated performance deviations from QASP standards, including justifications, impact assessment, and proposed mitigations.

5. Participating in a Monthly QASP Performance Meeting, as well as any other ad-hoc reviews that may arise during the monthly cycle, to review the Government's assessment of the Contractor's performance.
6. Preventing rework and ability to extend existing functionalities and features by planning and prioritizing appropriately.
7. Ensuring key work products, deliverables and ad-hoc tasks are completed on time and in accordance with established requirements, schedules, and quality standards.
8. Working with government stakeholders and GPMO to track and mitigate urgent defects as they occur.
9. Ensuring that all defects are managed and tracked in a timely manner, that is accepted by the stakeholders and GPMO.

3.1.2.4 Program Risk & Issue Management

The Contractor shall support the identification, documentation, communication, and mitigation of risks and issues that may affect program-level performance or outcomes. Program-level risks reported under this subsection shall be included in the Monthly Issue Risk & Issue Report.

The Contractor duties shall include, but are not limited to:

1. Identifying, documenting, and communicating to the GPMO project risks, mitigations, contingencies, and issues.
2. Supporting/participating in and responding to USCB risk management activities. This includes responding to information requests, supporting processes or procedures, maintaining and/or updating documentation, and attending both scheduled and ad-hoc meetings.
3. Capturing meeting notes and action items for program-level risk meetings that are jointly attended by federal and contractor staff and storing them in a centralized location that is accessible to all appropriate government and contractor management personnel.

3.1.2.5 Data Call and Audit Support

The Contractor shall facilitate the collection, analysis, and reporting of relevant data and information required for order audits or data calls.

The Contractor duties shall include, but are not limited to:

1. Supporting internal and external order audits on work performed under this order.
2. Supporting internal and external (e.g., Department of Commerce (DOC) & Government Accountability Office (GAO) inquiries) data calls related to work performed under this order.
3. Supporting the timely production and presentation of materials in support of Decennial Census Programs' data calls and oversight requests (e.g., GAO, Office of the Inspector General (OIG)).

3.1.2.6 Operational Readiness Reviews

The Contractor shall support Operational Readiness Reviews (ORR), which provides the Government confirmation that required capabilities, processes, and resources are prepared for upcoming operational phases or deployments.

The Contractor duties shall include, but not limited to:

1. Scheduling and conducting an ORR with Government stakeholders to present readiness evidence and obtain phase-gate approval.
2. Participating in business and technical interchange meetings and supporting data to ensure agenda items are fully covered.

3.1.2.7 Lessons Learned Management

The Contractor shall support the collection, analysis, documentation, and application of insights to improve processes, coordination, and performance throughout the life of the order.

The Contractor duties shall include, but are not limited to:

1. Executing a Lessons Learned Management process that captures insights, findings, and improvement opportunities derived from all order-administrative activities (e.g., onboarding, reporting, compliance, invoicing).
2. Capturing lessons learned throughout order execution.
3. Conducting periodic reviews, including at major program checkpoints, to analyze what worked well, what did not, and what changes are required to improve performance, processes, and coordination across Contractor and Government teams.
4. Incorporating lessons learned into updates of management artifacts, including the CMP, QCP, Staffing Plan, Subcontractor Management Plan, Communications Plan, Transition Plans, and relevant execution-level plans.
5. Maintaining a Lessons Learned Log that is version-controlled, and accessible to the Government.
6. Reporting lessons learned to the Government during regular meetings (weekly, monthly, quarterly), or as otherwise directed.
7. Supporting Government-led lessons learned activities, workshops, or after-action reviews by providing data, documentation, and personnel participation.

3.1.3 Transition Management

The Contractor shall support a smooth and efficient onboarding process at the start of the order and an orderly transition at its conclusion.

3.1.3.1 Transition-In

The Contractor shall support a smooth and efficient start of performance by establishing the personnel, processes, documentation, access, and resources necessary to initiate work under this order and to achieve full operational capability without disruption.

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, maintaining, and submitting a Transition-In Plan (TIP) for review and acceptance.
2. Executing the TIP, with continual tracking of transition activities, including but not limited to status of staff onboarding activities.
3. Participating in the transfer of all applicable knowledge, documentation, artifacts, and operational context from both the Government and, when applicable, from the outgoing Contractor, ensuring continuity of operations and retention of institutional knowledge.
4. Conducting ongoing Transition-In meetings with COR to report progress, address issues, and mitigate risks. Meeting frequency shall be coordinated with the COR based on transition needs.

3.1.3.2 Transition-Out

The Contractor shall support the orderly completion of work at the end of this order by ensuring the transfer of knowledge, data, artifacts, documentation, and responsibilities back to the Government, and when applicable, to an incoming Contractor, while maintaining continuity of operations, safeguarding Government information and assets, and closing out all order-related activities.

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, maintaining, and adhering to a Transition Out Plan (TOP), that ensures an orderly conclusion to the order and supports transfer of responsibilities to the Government and, when applicable, the incoming Contractor.
2. Coordinating and working in good faith with the Government and/or the incoming Contractor for an orderly and efficient transition of activities performed under this order.
3. Transferring all applicable knowledge, documentation, artifacts, and operational context to the Government and/or the incoming Contractor, ensuring continuity of operations and retention of institutional knowledge.
4. Providing all required close-out documentation, administrative records, and configuration items, including final deliverables, reports, and any other materials requested by the Government.
5. Return all Government-Furnished Equipment (GFE) and Government-Furnished Information (GFI) and ensure removal or secure disposition of Government data from Contractor systems.
6. Coordinating and implementing the TOP with the Government and/or incoming Contractor, providing timely and accurate information, data, and support for close-out reviews and audits.
7. Participating in a coordinated close-out effort to address any final Government concerns and confirming completion of all transition and close-out requirements.

3.1.4 Meetings and Reporting

The Contractor shall support consistent, transparent, and actionable communication of order status, risks, costs, schedule, staffing, and performance through structured meeting cadences and standardized reporting.

3.1.4.1 Kick-Off Meeting

The Contractor duties shall include, but are not limited to:

1. Coordinating and conducting a Kick-Off meeting with the Government at a mutually agreed time and location.
2. Providing meeting materials and status reports to allow for Government review and preparation.
3. Providing the Government with kick-off meeting minutes and an action items list from the meeting.

3.1.4.2 Weekly Status Report and Meeting

The Contractor duties shall include, but are not limited to:

1. Submitting a written weekly status report (WSR).
2. Conducting a status meeting to present the WSR.

3.1.4.3 Monthly Cost and Schedule Report and Meeting

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining a Monthly Cost and Schedule Report.
2. Conducting a cost and schedule meeting to present the cost and schedule report.

3.1.4.4 Program Management Review (PMR)

The Contractor duties shall include, but are not limited to:

1. Conducting Program Management Reviews (PMR).
2. Ensuring appropriate technical, operational, and management personnel from the Contractor's team attend PMRs to support discussion, answer questions, and resolve issues.
3. Producing a list of issues, decisions and action items identified during each PMR and submitting the meeting minutes and action item tracking list to the GPMO.

3.1.4.5 Monthly Risk and Issue Reporting

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining a Monthly Risk and Issue Report.

3.2 Task Area 2 – IT Project and Program Management

The Contractor shall provide IT Project and Program Management services that focus on the planning, tracking, management, and deployment of Decennial IT programs, projects, and initiatives including but not limited to program planning, governance, government staff training, organizational change management, and product delivery management. Task Area activities provide IT project and program management services for Decennial Census Programs, deliver IT training to federal staff, and implement organizational change management initiatives to prepare the workforce for technology modernization. Additionally, they manage product delivery

while ensuring smooth integration, collaboration, and review processes throughout the System Development Life Cycle (SDLC).

3.2.1 IT Project and Program Management Support

The Contractor shall provide IT Project and Program Management Support to effectively plan, execute, and monitor the implementation of IT projects and programs that align with the modernization and transformation objectives of the Decennial Census Program.

3.2.1.1 Project Management Support

The Contractor duties shall include, but are not limited to:

1. Implementing project management best practices following the System Development Lifecycle guidelines, including but not limited to budget estimation, scheduling, risk management, and change control.
2. Supporting the development, documentation, and maintenance of technical project documentation including, but not limited to:
 - a) Program/Project Charters
 - b) Business Cases
 - c) Project Plans
 - d) Business Continuity and Recovery Plans
 - e) Systems Management Plans (e.g., Test and Evaluation, Systems Engineering and Integration)
 - f) Risk Management documentation (e.g., risk management plans, program/project risk, issue, and opportunity registers, risk mitigation plans, risk contingency plans)
 - g) System Development Life Cycle documentation
 - h) Project Closeout Plans
 - i) Project post-mortems, after-action reviews, retrospectives, and Lessons Learned.
 - j) Executive and external stakeholder program/project presentations and Program Management Reviews (PMRs)
 - k) Status Progress Reports
 - l) Training documentation and training aids, as required, to provide proper operation, maintenance, and testing of processes, platforms, and systems
 - m) Schedule Management Plan
 - n) Change Control Plan
3. Reporting progress, issues, and risks.
4. Defining project milestone reviews to determine readiness to proceed with subsequent project activities.
5. Conducting and documenting periodic lessons learned, with metrics and recommendations for improvement.

6. Working collaboratively with stakeholders, other Contractors, and service / solution providers.
7. Establishing and documenting Contractor Points of Contact and defining interfaces between the project(s) and the Government

3.2.1.2 Agile Program Planning and Management Support

The Contractor duties shall include, but are not limited to:

1. Providing support to system teams and stakeholders with the implementation of agile methodologies and practices.
2. Preparing incremental, time-boxed program planning packages, procedures, and schedules to support coordinated delivery across multiple Agile teams.
3. Designing, developing, and implementing agile management processes, standards, and templates that support transparency and continuous improvement.
4. Creating directorate-specific solution delivery workflows and lean portfolio management approaches that enable iterative delivery and support a culture of continuous learning and improvement.
5. Providing a pool of certified agile practitioners with experience in large-scale, multi-team delivery environments.
6. Conducting and documenting periodic retrospectives and lessons learned with recommendations for improvement.
7. Supporting system teams with the transition to agile delivery practices, including but not limited to agile principles and values, scrum framework, sprint ceremonies, product backlog management, user story and acceptance criteria, and cross-functional collaboration.

3.2.1.3 IT Governance Support

The Contractor duties shall include, but are not limited to:

1. Supporting alignment of IT resources, scope, and schedules with Decennial Census Program objectives while managing risks. IT Governance structures include but are not limited to risk management board, change control board, program management governing boards, and Technical Review Boards (TRB).
2. Providing governance support services, including but not limited to preparing documentation (agendas, meeting notes, action items, etc.), facilitating meetings, and tracking action items to closure, in the support of technical governance processes and boards.
3. Developing and presenting technical briefings and recommendations.

3.2.1.4 Program Management Support

The Contractor duties shall include, but are not limited to:

1. Providing program management and planning to support the coordination and execution of Decennial Census Programs' projects, and activities, using agile and other development methodologies.

2. Assist with developing, documenting, maintaining, and updating IT strategic plans, program governance, and alternative analysis. This includes but is not limited to:
 - a) Assessing IT strategic plans, program governance, and program performance metrics and providing recommendations/strategies for improvement.
 - b) Supporting the analysis and evaluation of IT solutions and technologies.
 - c) Developing recommendations/strategies for integrating large scale IT systems.
3. Establishing and implement processes, templates, and tools to support the initiation, planning, requirements, design, development, integration, implementation, and monitoring of system development life cycle functions.
4. Leveraging Decennial GFE program / project management tools.
5. Develop an approach to collecting lessons learned that allows various methods of feedback such as facilitated sessions, surveys, focus groups, etc.
6. Developing, documenting, and maintaining, Lessons Learned Reports to identify areas for improvement in Decennial operations and summarize the key takeaways.

3.2.1.5 Schedule Management Support

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, maintaining, providing status of, and management of project schedules across programs that map to the Decennial work schedule.
2. Supporting execution and facilitation of the schedule management processes.
3. Creating the IT program and project level schedule templates for incorporation into the Integrated Master Schedule (IMS) or other Decennial Census Program schedules.
4. Providing status, inputs, and metrics on IT tasks in the IMS and residual project schedules.
5. Identifying any risks, issues or dependencies which may cause project schedule delays and provide recommendations, mitigation strategies, and contingency planning.
6. Monitoring program and project schedules so that all programs and projects meet milestones.

3.2.1.6 Reporting Support

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining a reporting strategy and plan.
2. Assisting in monitoring and reporting the progress of achieving security status including but not limited to Authority to Operate (ATO) and penetration testing.
3. Collecting, analyzing, and reporting metrics.
4. Developing, documenting, and disseminating status reports and visualizations to USCB stakeholders.

3.2.1.7 Risk Management Support

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and executing the Decennial Census Program's risk management processes, including facilitating the identification, tracking, and reporting of program and project risks.
2. Facilitating and executing the risk, issues, and opportunity management process including, but not limited to, providing technical risk, issue, and opportunity inputs including assessment, monitoring, and contingency/mitigation/avoidance planning.
3. Updating and maintaining technical risks in the Decennial Census Program Risk Register.
4. Coordinating the resolution of Decennial Census Program and/or project level risks/issues.
5. Providing technical insight into project risks and ongoing performance of projects.
6. Coordinating with other programs on risk management, where applicable.
7. Coordinating with Decennial SoS teams on risk and issue management,
8. Facilitating the development and implementation of contingency and mitigation plans.

3.2.1.8 Change Control Management Support

The Contractor duties shall include, but are not limited to:

1. Providing support for change control management including but not limited to participation, information collection, and feedback for Change Control Boards and TRBs.
2. Facilitating change control activities and impact analysis for project and program-level change requests while following the established change management processes and standards.
3. Assessing the pre-existing Change Control process and providing improvement and recommendations (e.g., people, processes, technology).
4. Validating Change Request implementation across the Decennial Census Program and projects.
5. Designing, developing, and implementing change and configuration management processes, standards, and templates to minimize the likelihood of disruptions, unauthorized alterations, and errors.
6. Conducting and documenting periodic lessons learned, with recommendations for improvement.
7. Reporting on change control activities and progress.

3.2.1.9 Disposition Management Support

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, implementing, and maintaining the Decennial Disposition Management Plan according to schedule.
2. Assessing systems and their corresponding capabilities including but not limited to infrastructure, applications, middleware, network, code, and tools along with current and future USCB needs to determine their approved disposition target state along with the rationale.

3. Planning for, assisting, and tracking the archival or transfer of data, Decennial SoS components, and system development life cycle artifacts.
4. Coordinating and communicating disposition and closeout activities with stakeholders and SoS teams.
5. Conducting and documenting periodic lessons learned, with recommendations for improvement.
6. Reporting on disposition management activities and progress.

3.2.1.10 Technical Writing and Editing Support

The Contractor duties shall include, but are not limited to:

1. Creating, updating, and maintaining technical writing documentation and graphic illustrations, including but not limited to Contractor deliverables, work products, reports, presentation materials, metrics, and visualizations.
2. Adhering to the USCB documentation standards and best practices.

3.2.1.11 Communication Strategy and Planning Support

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining communication plans and coordinating with other stakeholders as needed.
2. Developing, documenting, implementing, and maintaining a standard communication delivery method, channel, and content.
3. Developing, documenting, and maintaining stakeholder mapping.
4. Developing, documenting, implementing, and maintaining processes to receive, review, and incorporate (as applicable) stakeholder feedback on the communication strategy and implementation.

3.2.2 Government Staff Training

The Contractor shall provide training services to Government Staff to facilitate a smooth transition and maximize the value and impact of the implemented technologies for the USCB.

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining a skills assessment of Government staff.
2. Identifying gaps between organizational skills needed compared to the current skills assessment.
3. Performing a current state assessment of training needs to meet Decennial Census Programs' transformation and application modernization needs.
4. Developing, documenting, maintaining, and updating training plans for Government staff.
5. Developing and delivering training curriculum.
6. Executing and tracking staff training.
7. Assessing the effectiveness of the training plan.

8. Improving training through continuous feedback.

3.2.3 Organizational Change Management

The Contractor shall provide Organizational Change Management services to effectively plan, implement, and manage changes within USCB and to ensure successful adoption and transition.

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, updating, and executing a knowledge transfer strategy and organizational change management plans for federal resources
2. Providing facilitated workshops to support Decennial applicable enterprise and technology transformation initiatives.
3. Developing, documenting, and implementing a training and organizational change management repository.
4. Developing and implementing a communication plan to support technology transformation and application modernization for the Decennial Directorate.
5. Designing, engineering, developing, and executing an integrated approach to knowledge management / information management.
6. Collecting and reporting periodic lessons learned, including organizational change management metrics.

3.2.4 Product Delivery Management

The Contractor shall provide Product Delivery Management services to support the planning, tracking, logistics, and deployment of systems and processes.

3.2.4.1 Operational Delivery Management

The Contractor duties shall include, but are not limited to:

1. Providing IT and non-IT delivery management services including but not limited to planning, tracking, logistics, management, and deploying of various components, systems, and processes.
2. Updating and implementing changes to the Operational Delivery Management Plan (ODMP).
3. Establishing Operation Delivery (OD) teams for each operational delivery.
4. Facilitating OD meetings with an agreed to agenda.
5. Capturing and disseminating meeting minutes and action items related to integration and implementation of operational delivery.
6. Monitoring and managing integration points, interdependencies, and milestones among stakeholders and service / solution providers for product delivery and collaborating to resolve blockers.
7. Investigating and tracking integration and implementation action items to resolution.

8. Developing and providing executive level briefings and presentation materials for OD, highlighting systems at risk for meeting key Integration and 2030 Census Integration and Implementation Plan (IIP) milestones dates.
9. Monitoring and managing integration points, milestones, and interdependencies among the Decennial Census Program SoS service providers (e.g., contracts, enterprise program).
10. Identifying, documenting, mitigating, and managing OD integration risks and escalating, as necessary.
11. Supporting operational delivery starting from the IBR through closeout and lessons learned.
12. Conducting periodic lessons learned.
13. Collecting and reporting on OD metrics.
14. Conducting phase gate reviews and developing detailed runbooks for software and feature releases into production.

3.2.4.2 Readiness Reviews (RRs)

The Contractor duties shall include, but are not limited to:

1. Supporting and facilitating applicable RR and reporting.
2. Developing, documenting, and maintaining RR materials.
3. Capturing and tracking actions (ensuring all actions have an assignee, have a due date, and have been completed) and escalating and resolving all actions from the RR.
4. Producing summary reports with metrics.
5. Collecting RRs entrance and exit criteria for all systems as identified in the Decennial Census Program Master Entrance / Exit Criteria List prior to each RR.
6. Reviewing and validating all RR entrance and exit criteria as identified in the Decennial Census Program Master Entrance/Exit Criteria List prior to the RR per OD and provide recommendations.

3.2.4.3 Runbooks

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, developing, and maintaining detailed Runbooks. The activities associated with runbook support include but are not limited to:
 - a) Selecting users to participate and coordinate with stakeholders and solution providers.
 - b) Monitoring system performance.
 - c) Providing a venue for feedback based on predefined success criteria.
2. Facilitating Runbook walkthroughs with service providers and stakeholders.
3. Monitoring and reporting the status of execution of Runbook activities.

3.2.5 System Level Testing

The Contractor shall provide System Level Testing services to ensure that integrated systems, applications, and services operate reliably, securely, and as intended across the full program ecosystem.

The Contractor duties shall include, but are not limited to:

1. Supporting system-level Integration testing to demonstrate the feasibility of conceptual approaches, evaluating design risk, identifying alternatives, comparing and analyzing trade-offs, and assessing satisfaction of solution requirements.
2. Instituting a continuous evaluation of the testing process after each integration test cycle, and after an agreed number of Program Increments.
3. Performing automated testing (unit and project level). Please refer to Section 3.4.3 (Test Execution) for additional information and requirements pertaining to Testing support.
4. Develop, update, maintain, and disseminate a System Defect Management Plan.
5. Collaborate with system and/ or component owners on defect resolution including, but not limited to, categorizing, tracking, and dispositioning all software defects.
6. Develop, update, maintain, and disseminate Program Test Analysis Reports that reflect testing outcomes across the entire system or program ecosystem.
7. Create integrated test data to support system-level testing across multiple applications, services, and platforms.
8. Conduct multi-layer testing (including the validation of outputs) across various platforms (e.g., smartphones, laptops, tablets, and desktops) to validate consistent system behavior.
9. Conduct program level testing in the fully integrated program ecosystem environment, validating end-to-end functionality, data flow, and performance across all relevant systems and external interfaces.
10. Support and participate in reviews such as test readiness and test analysis reviews.
11. Develop and maintain system-level test cases, test scripts, test scenarios, and test metrics that reflect cross-system requirements.
12. Maintain source code and documentation repositories relevant to system level integration and testing activities.
13. Coordinate system-level testing activities across teams, environments, and stakeholders to provide alignment, scheduling, and resource availability for integrated test efforts.

3.2.6 Test Coordination

The Contractor shall provide Test Coordination services to plan, align, and synchronize testing activities across multiple solutions, programs, and stakeholders.

The Contractor duties shall include, but are not limited to:

1. Providing coordination between various solution providers, testing areas and program teams to ensure cohesive and comprehensive testing efforts across multiple projects.

2. Working closely with other program and project managers to develop and align an approach for integrated testing, ensuring that all interdependencies and cross-program impacts are considered.
3. Formulating and communicating testing strategies that encompass the full scope of interconnected solutions, testing areas and programs, addressing potential integration points, shared resources, and common objectives.
4. Ensuring effective communication and collaboration between solution teams, testing areas and program teams.
5. Identifying and resolving any issues related to the integration of systems and components across programs.

3.3 Task Area 3 – Architecture and Engineering

The Contractor shall provide Architecture and Engineering services to define needs and functions early in the development cycle. This entails defining and documenting all requirements at the start, then generating and validating the design with the entire product cycle in mind – operations, performance, testing, cost and scheduling, training and support, security, and disposal. Task Area 3 activities deliver systems architecture and engineering services inclusive of all phases of the systems' development and operational lifecycle.

3.3.1 System of Systems Architecture and Engineering

The Contractor shall provide Systems of Systems (SoS) Architecture and Engineering services to support the design and development of optimized application architectures and data repositories.

The Contractor duties shall include, but are not limited to:

1. Reviewing and assessing current Decennial and enterprise systems architecture and engineering footprints, plans, standards, and future roadmap of on-prem, Software as a Service (SaaS), cloud / multi-cloud architectures, practices, technologies, tools, services, capabilities, and perform a gap analysis.
2. Developing and documenting the gap analysis findings and provide recommendations.
3. Implementing, maintaining, and supporting the recommended activities per the gap analysis.
4. Reviewing current schedules, requirements, roadmaps, and other planning artifacts.
5. Developing, documenting, and maintaining a system architecture and engineering roadmap and associated schedules.
6. Establishing, documenting, and maintaining systems architecture and engineering standards, processes, and procedures for SoS service providers based on best practices suitable for the Decennial Census Program.
7. Communicating the standards, processes, and procedures for architecture, infrastructure, and SoS service providers.
8. Establishing, documenting, and maintaining coding standards for the purposes of auditing and logging these standards.

9. Developing, documenting, implementing, and maintaining Decennial systems engineering plans including, but not limited to, a Systems Engineering Management Plan, Systems Engineering Plan, and Software Developer Guide.
10. Collaborating with internal and external service providers, solution providers and third-party vendors to support implementation of the systems architecture and engineering processes and procedures.
11. Developing, documenting, maintaining, and providing periodic reports, status updates, and data visualizations regarding system architecture and engineering status and activities.
12. Establishing, updating, and maintaining systems architecture and engineering artifacts including but not limited to documentation, work products, and deliverables, in alignment with USCB Enterprise Architecture.
13. Support program risk management by identifying, documenting, analyzing, and managing risks and issues that could impact scope, cost, or schedule.
14. Creating and maintaining a Decennial Census Program Interface Catalog with all interface definitions and detail.
15. Developing, documenting, and maintaining the Decennial Census Program's Line of Business Architecture documents.
16. Developing, documenting, and maintaining the Decennial SoS architecture.
17. Establishing, documenting, maintaining, and distributing workflow diagrams.
18. Supporting the development of Decennial mobile standards to guide mobile development efforts in accordance with any existing enterprise-level standards.
19. Identify business threads that represent the various sequence of activities that are dependent on each other and collectively contribute to achieving specific objectives or outcomes needed to create system architecture and engineering artifacts.
20. Planning, coordinating, executing, and/or participating in ongoing architecture and engineering reviews of systems, programs, infrastructure, and supporting services/capabilities to verify that they are properly engineered and developed in a fashion that is in alignment with architecture and engineering standards and guidelines, and provide best practice recommendations as appropriate.

3.3.1.1 Application Rationalization

The Contractor duties shall include, but are not limited to:

1. Reviewing and assessing the current state of application and system architectures, inventory, infrastructure, capabilities, and services for the target states of the applications.
2. Designing, developing, implementing, maintaining, and supporting an application rationalization and system assessment plan from current/intended states to target states based on established application architecture standards, templates, and patterns, that meet Decennial Census Program's schedules, plans, roadmaps, and requirements.
3. Designing, developing, and maintaining decision principles for migration, informed by application and team readiness, business priorities, application modernization strategy, and vendor capabilities.

4. Designing, developing, documenting, implementing, and maintaining an application rationalization plan roadmap and schedule.
5. Implementing continuous improvement on the Decennial application rationalization and system assessment plan while leveraging lessons learned from detailed application rationalization efforts.
6. Leading, supporting, and facilitating application rationalization discussions, meetings, work sessions, and collaborations, with federal and contract resources, third parties, Engineering Working Group (EWG), and other stakeholders.

3.3.2 Cloud Architecture and Engineering

The Contractor shall provide Cloud Architecture and Engineering services to support the design and implementation of cloud-based solutions that align with the Decennial Census Program's goals.

The Contractor duties shall include, but are not limited to:

1. Reviewing and assessing current cloud architecture and engineering plans, practices, and procedures, and making recommendations for optimizing, migrating, and modernizing cloud applications.
2. Establishing, documenting, implementing, and maintaining cloud naming standards (i.e., for naming workloads, instances, etc.) and incorporating names into resource tagging.
3. Establishing, documenting, implementing, and maintaining data tagging strategies and standards to help manage, search for, and filter resources as well as maintain cost traceability.
4. Establishing, documenting, implementing, and maintaining software data validation rules and standards to meet data quality, compliance, and governance needs.
5. Establishing, documenting, implementing, and maintaining cloud messaging standards to provide secure, reliable program-to-program communication including, but not limited to, simple notifications, pub/sub implementation, etc.
6. Establishing, documenting, implementing, and maintaining cloud development standards in adherence to the Secure Software Development Life Cycle process, depending on the application's requirements.
7. Establishing, documenting, implementing, and maintaining design-patterns to be employed across cloud-native applications.
8. Establishing, documenting, implementing, and maintaining a cloud control plane to provide management across cloud application deployment.
9. Planning, coordinating, and executing ongoing reviews and gap analyses of cloud native processes, policies, and procedures, and making recommendations based on best practices as appropriate.
10. Reviewing and analyzing cloud security tools and implementations as well as making recommendations based on Decennial security requirements and best practices.
11. Establishing, documenting, and maintaining a roadmap of required cloud native services, capabilities, and requirements based on Decennial Census Program, security, and operational milestones and requirements, and making recommendations for development, implementation, enhancement, and maintenance of these services.

12. Establishing, documenting, implementing, and maintaining automation capabilities and enhancements for Decennial and Decennial-supporting systems, application, services, and capabilities.

3.3.3 Systems Engineering Management

The Contractor shall provide Systems Engineering Management services to effectively plan, coordinate, and oversee the development and implementation of new systems and technologies.

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, and maintaining systems engineering management process documentation including but not limited to System Engineering Management Plan (SEMP), Test and Evaluation Management Plans (TEMP), Standard Operating Procedures, Decennial Census Program Defect Management Plan, Decennial Census Program Disposition Management Plan, the ODMP, and Concept of Operations.
2. Supporting technical integration within all functional areas within the Operational Delivery and integrated product teams to provide consistent systems engineering processes and the execution of those processes in accordance with the SEMP.
3. Distributing and communicating updates made to the systems engineering processes with all impacted stakeholders and managing the transition to, and adoption of, the updated processes.
4. Developing, documenting, and maintaining IT planning and IT strategy documents, such as the IT Strategy and Roadmap.
5. Creating and maintaining the IIP for Decennial Census Program including but not limited to defining:
 - a) Major Milestones (RR) and success criteria.
 - b) Interim Milestones (Milestones leading to RR).
 - c) Narrative plan identifying how to integrate the solutions within the SoS to get to production.
6. Validating system engineering designs and plans and support the implementation of the Decennial Census Program Disposition Management Plan.
7. Facilitating and supporting TRBs as required.
8. Facilitating, coordinating, and overseeing technical and business integration activities between Decennial, various Census stakeholders, systems, and applications including, but not limited to, Business Ecosystem (BE), third-party contractors, purpose-built systems, and other applications to the SoS.
9. Developing, documenting, implementing, maintaining, and supporting a technical onboarding process and training for systems supporting the Decennial Census Program.
10. Assisting Integrated Product Teams, OD Teams, and system teams, in implementing System Development Life Cycle tools.

3.3.4 Requirements Management

The Contractor shall provide Requirements Management services to effectively capture, analyze, and manage requirements for the development and implementation of the Decennial Census Program transformation and modernization solution.

3.3.4.1 Requirements Development

The Contractor duties shall include, but are not limited to:

1. Eliciting, collecting, documenting, managing, facilitating, analyzing, validating, and allocating Decennial Census Program requirements to meet industry best practices such as specific, measurable, achievable, relevant, and time-bound (SMART.)
2. Developing and maintaining a requirements repository.
3. Creating and maintaining a capabilities/features catalog.
4. Creating and maintaining a requirements management plan.
5. Collaborating with and providing assistance to the relevant stakeholders, SMEs, and service/solution providers in the development of functional, non-functional, and domain requirements including, but not limited to, performance, user, regulatory, and compliance requirements and Decennial Census Program/Project detailed Business requirements.
6. Facilitating joint application development sessions to identify and document requirements for the program.
7. Analyzing, organizing, prioritizing, and verifying solution and stakeholder requirements for completeness and coverage of desired business capabilities, and delivering recommendations.
8. Identifying requirement gaps inclusive of cross-operational requirements.
9. Developing and establishing processes, templates, and tools necessary to develop and manage Decennial Census Program/Project requirements.
10. Collaborating with relevant stakeholders to create, document, maintain, and disseminate wireframe diagrams using the Unified Modeling Language.
11. Developing and documenting detailed requirements and acceptance criteria using Incremental, Iterative, Agile, and Waterfall development methodologies, consistent with the Decennial Census Program Requirements Management Plan.
12. Developing, documenting, and maintaining requirements reporting metrics for applications developed and across SoS.
13. Assisting the Decennial service/solution providers in the implementation of the requirements management tool and status reports.
14. Collaborating with relevant stakeholders in creating requirements baselines.
15. Performing a requirement change impact analysis and delivering recommendations, as necessary.
16. Communicating and supporting change control for the requirements baselines over the Decennial Census Programs and Project lifecycles.

3.3.4.2 Requirements Traceability

The Contractor duties shall include, but are not limited to:

1. Maintaining and managing the traceability of requirements.
2. Validating end-to-end requirements traceability across all levels of requirements and testing.
3. Developing and maintaining the Requirements Traceability Matrices and maintaining requirements traceability throughout the Systems Development Life Cycle.

3.3.5 Facilitation Services

The Contractor shall provide Facilitation Services to support efficient communication and coordination among stakeholders involved in the Decennial Census Program's modernization efforts.

The Contractor duties shall include, but are not limited to:

1. Organizing, designing, leading, and participating in meetings, working sessions, troubleshooting sessions, and other collaborative sessions with solution/service providers, federal and contract staff, third parties, and other stakeholders as necessary.
2. Capturing and/or scribing session meeting notes, decisions/recommendations, and action items.
3. Tracking action items to resolution and escalate any blockers.
4. Organizing, designing, leading, and participating in follow-on business and technical collaborative sessions required to resolve any outstanding issues, problems, or action items.

3.4 Task Area 4 – Application Development and Management Services

The Contractor shall provide Application Development and Management Services to support application design and development, architecture, testing, security, operations and maintenance, cloud application migration, research and development, and innovation. This includes supporting the development, modernization, and management of applications using cloud native best practices to provide the highest levels of security, quality, and speed.

3.4.1 Application Design and Development

The Contractor shall provide Application Design and Development services to support the design and development of new applications that align with the Decennial Census Program's objectives and leverage modern technologies and best practices.

The Contractor duties shall include, but are not limited to:

1. Designing and developing applications, including but not limited to the following elements and/or factors:
 - a) Application code (purpose-built custom, commercial-off-the-shelf (COTS), Government off the shelf (GOTS), low-code/no-code, SaaS, open-source)
 - b) Databases and data structures

- c) User interfaces (UI)
 - d) User experience (UX)
 - e) Wireframes and prototypes
 - f) Application security (such as encryption, Multi-Factor Authentication, authentication, and authorization)
 - g) For scalability and application performance
 - h) Error handling and logging
 - i) Continuous Integration / Continuous Delivery (CI/CD) integration
 - j) Data quality assurance
 - k) 508 compliance
 - l) Deployment strategies (code and infrastructure)
 - m) Microservices
 - n) Containerization
 - o) Cloud-native applications
 - p) Back-end and front-end development
 - q) Serverless architecture
 - r) Business intelligence
 - s) Advanced Data Science, including AI/ML, and Robotic Process Automation
2. Managing, implementing, and maintaining, at a minimum:
 - a) Version control
 - b) Quality assurance
 - c) Technical documentation
 - d) Deployment strategies (code and infrastructure)
 - e) Requirements analysis
 - f) Use-case modeling
 - g) Scaled agile framework
 - h) Business intelligence
 3. Following System Development Life Cycle methodology.
 4. Incorporating functional and non-functional requirements into design.
 5. Documenting and formalizing application design and development industry best practices so they can be incorporated and/or adopted.
 6. Performing and supporting testing necessary for application development.
 7. Performing and supporting data migration and system integration activities.
 8. Leading, supporting, and participating in code reviews/audits.

3.4.1.1 DevSecOps Methodology

The Contractor duties shall include, but are not limited to:

1. Utilizing a DevSecOps methodology as the preferred approach for integrating security into development and operations
2. Updating and maintaining DevSecOps standards and best practices.
3. Creating and maintaining DevSecOps documentation, procedures, and policies.
4. Documenting and maintaining Continuous Integration/Continuous Delivery (CI/CD) standards to automate the deployment process and provide rapid and reliable releases.

3.4.1.2 DevSecOps Engineering

The Contractor duties shall include, but are not limited to:

1. Implementing DevSecOps standards and processes based on Decennial DevSecOps methodology.
2. Implementing CI/CD to automate the deployment process and provide rapid and reliable releases.
3. Working in collaboration with the application, security, and infrastructure teams in the implementation of CI/CD pipelines using DevSecOps principles and best practices for code development and deployment.
4. Creating and/or adhering to DevSecOps security gates for all environments used for the development and promotion of software and application code.

3.4.1.3 DevSecOps Processes and Tools (CI/CD)

The Contractor duties shall include, but are not limited to:

1. Utilizing DevSecOps processes and tools for the promotion of applications through all USCB environments.
2. Installing/deploying, configuring, and administering DevSecOps tools following Decennial change management procedures.
3. Developing, documenting, implementing, and maintaining the DevSecOps CI/CD solution, process, and software tools for engineering activities, including but not limited to:
 - a) Manage Source Code Deployment and Configuration Files
 - b) Develop Source Code
 - c) Dynamic / Static Code Analysis
 - d) Optimize Source Code (code quality)
 - e) Build Application or Service
 - f) Secure API use
 - g) Create and Perform Automated Testing
 - h) Database change and management

- i) Manage Application and Service Artifacts
 - j) Manage Test Data
 - k) Orchestrate End to End Release
 - l) Manage Deployment Database
 - m) Code Signing and Integrity Checks
 - n) Apply Feature Toggling
 - o) Policy enforcement points and policy decision points (PEP and PDP)
 - p) Code-merge and code reviews
 - q) Supply chain code analysis (software bill of materials (SBOM) compliance)
4. Accommodating existing release methodologies such as manual deployment or shell script for CI/CD.

3.4.1.4 Section 508 Compliance

The Contractor duties shall include, but are not limited to:

1. Following all Section 508 compliance rules.
2. Collaborating with the USCB's Section 508 Compliance office to assess what compliance level is needed, designing and developing applications to that level, and remedying the application to the required testing level.

3.4.1.5 Software Application Administration

The Contractor duties shall include, but are not limited to:

1. Developing, installing, configuring, securing, administering, and managing Software/Services, application hosting platforms/services, cloud services, cloud storage, serverless platforms and its supporting third-party add-ons. Application types include but are not limited to purpose-built custom, COTS, GOTS, geospatial, low-code/no-code, SaaS, and open-source.
2. Designing, documenting, implementing, maintaining, and supporting a software configuration and release management process in alignment with System Development Life Cycle requirements.
3. Establishing, documenting, and maintaining software configuration management documentation.
4. Designing, developing, documenting, and maintaining automated CM workflows.
5. Developing, documenting, implementing, and maintaining secure baselines for all systems/software.
6. Troubleshooting software installations, secure configurations, and performance problems.
7. Validating applications are installed, configured, and their use is consistent with license agreements.
8. Developing systems and services to ensure they are highly available and fault tolerant.

9. Monitoring, performance tuning, and troubleshooting support.
10. Implementing and supporting disaster recovery, continuity of operations (COOP), backup, and recovery per the USCB guidance.

3.4.2 Application Architecture

The Contractor shall provide Application Architecture services to support designing and defining the overall structure and framework of the applications within the Decennial Census Program.

3.4.2.1 Application Architecture Support

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, and maintaining application architecture standards, patterns, and types that prioritize design factors such as, but not limited to, interoperability, reliability, availability, performance, scalability, extensibility, security, and other DevSecOps dimensions, for applications within the Decennial IT solutions architecture.
2. Establishing, documenting, and maintaining tools, processes, documentation, and templates required to develop and manage the Application Architecture.
3. Performing business and solution architecture activities to appropriately design and model requirements in alignment with the Decennial architecture goals.
4. Creating and maintaining application architecture artifacts, documents, presentations, and brief governance boards, as requested by the Government.
5. Performing reviews and assessments of current and proposed application architecture and providing recommendations that meet business needs.
6. Developing application architecture that supports programs in the Decennial Census Program that are in alignment and compliant with the programs' business and operations plans, schedules, documentation, and governance.
7. Developing, documenting, and maintaining the interconnection and interface design and workflow artifacts.
8. Describing the behavior and structure of applications and how they interact with each other and with users.
9. Developing, documenting, and maintaining Concepts of Operations based on business requirements by application.
10. Leveraging enterprise architecture (EA) tools, models, and principles to create Decennial Census Program application architecture artifacts.
11. Participating in EWG reviews of application architecture and other governance boards, as required, and providing technical recommendations and guidance that meet business needs.

3.4.2.2 Cloud Application Architectures

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining secure multi-cloud application architectures whenever possible that utilize cloud services, methodologies, and

products such as, but not limited to, microservices, infrastructure as code, serverless computing, and containerization.

2. Providing and utilizing application architecture best practices and solutions to enable the development and migration of the Decennial application portfolio.

3.4.2.3 Application Architecture Trade-off Analysis

The Contractor duties shall include, but are not limited to:

1. Developing alternative application architectures that support the application's functional and non-functional requirements.
2. Developing, documenting, and disseminating criteria for trade-off analysis.
3. Performing trade-off analysis of application architectures to determine the optimal solution based on Decennial's requirements.
4. Providing recommendations for best fit solutions to the Government based on the trade-off analysis.

3.4.2.4 Application Programming Interfaces (API) Management (Discovery, Security, Governance)

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining API standards and best practices for the Decennial IT Solutions Architecture (example: Coarse API to granular API approach).
2. Developing, documenting, and maintaining API and their related components, including the Business Rule Engines.
3. Implementing application lifecycle management practices and tooling that align with any Zero Trust Architecture (ZTA) requirements and with API security requirements.

3.4.2.5 Application Financial and Operational Principal (FinOps) Architecture

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining application architectures consistent with Decennial FINOPS procedures.
2. Conducting periodic cost assessments and making recommendations to reduce costs and optimize performance based on business needs.
3. Developing, documenting, implementing, and maintaining automated FINOPS monitoring and alerting capabilities to control cost and optimize performance based on triggers and thresholds; generate reports, implement automated remediation activities, perform budget forecasting, and establish predictive cost models using Census-provided FINOPS tools.

3.4.3 Application Testing

The Contractor shall provide Application Testing services to ensure the quality, reliability, and proper functionality of applications and software being developed or modernized, including integration testing, to support effective quality control and quality assurance.

3.4.3.1 Multi-layer Testing

The Contractor duties shall include, but are not limited to:

1. Conduct the following types of testing, as requested, including, but not limited to:
 - a) Functional Testing
 - b) Non-functional Testing
 - c) Interface Testing
 - d) System Testing
 - e) Performance Testing
 - f) Scalability Testing
 - g) Load / Stress Testing
 - h) Usability Testing
 - i) User Acceptance Testing
 - j) Regression Testing
 - k) Security Testing (Pen Testing / Penetration Testing)
 - l) Section 508 of the Rehabilitation Act of 1973 Testing*
 - m) Pair-wise Testing
 - n) Soak Testing
 - o) Path Testing
 - p) Exception Testing

**Note: The USCB Office of the Chief Information Officer (CIO) will provide independent Section 508 certification.*

2. Support software testing in Incremental, Iterative, Agile, and Waterfall development methodologies.
3. Participate in and support testing.

3.4.3.2 Test Planning

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining test strategies for testing applications.
2. Developing, documenting, and maintaining test plans and schedules in alignment with Decennial schedules.
3. Developing, documenting, implementing, and maintaining automated test capabilities, to the maximum extent possible.

4. Collaborating with stakeholders to review, assess, and validate application architecture documents, artifacts, and application requirements with the goal of having well-defined and accepted test requirements.
5. Briefing stakeholders on test plans and schedules.
6. Participating in requirements gathering, workshops, user meetings/discussions, and walkthroughs.
7. Coordinating with infrastructure team on test infrastructure resources needed.

3.4.3.3 Test Data and Scenario Creation

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, and maintaining a test data strategy and plan in accordance with Census and Decennial data security policies.
2. Creating and maintaining synthetic test data to support software testing.
3. Creating and maintaining datasets from sensitive data to support software testing.
4. Developing, documenting, and maintaining test cases, test scripts, test scenarios and automating to the maximum extent possible.
5. Developing, documenting, implementing, and maintaining a test management approach that supports requirements traceability, defect injection rate analysis, and retrospectives throughout the software delivery pipeline.
6. Deploying tooling and adopting coding practices that enable automated test frameworks including, but not limited to, functional unit testing and non-functional performance testing.

3.4.3.4 Test Execution

The Contractor duties shall include, but are not limited to:

1. Executing test cases and providing traceability.
2. Supporting software testing in Incremental, Iterative, Agile, and Waterfall development methodologies.
3. Conducting testing using Government-provided tools and equipment.
4. Conducting multi-layer testing, including the validation of outputs, on multiple platforms, as required (e.g., mobile devices, laptops, tablets, and desktops).
5. Conducting application testing in the integrated environment across all relevant hosted environments.
6. Providing testing approaches that are highly configurable.
7. Conducting automated software testing supported by screen-recording tools, shell scripting, synthetic data generation, etc.
8. Supporting and participating in reviews including, but not limited to:
 - a) Technical reviews such as code reviews, test readiness and test analysis reviews.
9. Collaborating with stakeholders and application developers on defect resolution.

3.4.3.5 Test Reporting

The Contractor duties shall include, but are not limited to:

1. Developing, updating, maintaining, and disseminating Test Analysis Reports.
2. Categorizing, tracking, and dispositioning all software defects (can include severity level or type of defect, etc.).
3. Providing reporting metrics and data visualizations that capture items including, but not limited to, defects, testing trends, timeframes, and categories.

3.4.4 Application Security

The Contractor shall provide Application Security services to support the protection and integrity of applications against potential threats and vulnerabilities.

The Contractor duties shall include, but are not limited to:

1. Implementing documented policies and procedures to comply with the USCB's security policies.
2. Building, integrating, and automating with security in mind to protect sensitive data and applications while following guidance, regulations and policies provided by the Office of Information Security (OIS).
3. Developing and maintaining metrics and reporting dashboards to track progress and improvement of application security posture over time.
4. Supporting Decennial Census Program ATO processes, continuous monitoring, contingency planning, and disaster recovery.
5. Implementing Zero Trust (ZT) principles for applications, workload, and data.
6. Evaluating newly approved technologies and implementing supplemental controls that are suitable for the Decennial Census Program use cases.
7. Adapting to changing deployment models and attack surfaces.
8. Assisting in the planning and designing of application and software security architecture that can reduce the likelihood of attack.
9. Supporting the OIS security assessments and ATO processes for systems and applications.
10. Supporting the development and management of Interagency Security Agreements (ISA). This includes but is not limited to:
 - a) Compiling and maintaining a list of ISAs.
 - b) Evaluating, coordinating, and updating ISAs so documentation is current.
 - c) Developing a process for tracking ISAs so they are in alignment with the Bureau's security policies and the system development life cycle.
11. Documenting and implementing hardening techniques including, but not limited to, workload protection, container image scanning, private registries/repositories, and baselines during build time and run time.
12. Generating Software Bill of Materials (SBOMs) in accordance with Federal guidance (Office of Management and Budget (OMB), Department of Homeland Security,

Cybersecurity and Infrastructure Security Agency, and National Telecommunications and Information Administration, regulations, Executive Orders, and USCB implementation:

- a) Utilize a Census-provided tool to continuously monitor and report compliance.
 - b) Document and track baseline information.
13. Collaborating with service providers and the OIS on Decennial security requirements to document and implement security standards for systems, application, and data.
 14. Assisting in the development and maintenance of application security playbooks related to secure coding, security integration, security operations, and incident response.

3.4.4.1 Zero Trust

The Contractor duties shall include, but are not limited to:

1. Establishing a robust and proactive security posture that mitigates the risk of unauthorized access, data breaches, and lateral movement within the network, ultimately enhancing the overall security and resilience of the systems and applications that are developed and managed.
2. At a minimum, utilizing ZT cybersecurity framework and assume that no user or device will be automatically trusted regardless of their location or network connection.
3. Adhering to the most updated federal guidelines and mandates. These include but are not limited to:
 - a) Executive Order (EO) 14028 "Improving the Nations Cybersecurity".
 - b) OMB Memorandum M-22-09 "Moving the U.S. Government Toward Zero Trust Cyber Security Principles".
 - c) National Institute of Standards and Technology (NIST) Special Publication (SP) 800-27: Zero Trust Architecture (ZTA).
4. Conducting gap analysis and making recommendations on ZTA implementations supporting application development.
5. Implementing recommendations from ZTA gap analysis.
6. Developing, documenting, implementing, and maintaining ZTA domains applicable to Decennial application development.

3.4.4.2 Security Architecture

The Contractor duties shall include, but are not limited to:

1. Incorporating IT security compliance, principles, and best practices during the design of applications.
2. Performing a pre-assessment of applicable security controls for newly developed and/or modified software.
3. Implementing security controls in accordance with defined control requirements and remediate identified vulnerability and compliance findings.
4. Providing written recommendations for meeting National Institute of Standards and Technology (NIST) Cybersecurity Framework and the NIST 800-53x based on

application changes, baseline changes, or security program changes, as applicable and in accordance with the USCB security program.

5. Establishing, documenting, implementing, and reporting metrics to measure the effectiveness of application security.
6. Implementing and applying secure software design principles and patterns, ensuring that security is incorporated into the architecture from the early stages of development.
7. Implementing secure integration and API design practices, including authentication and authorization mechanisms, input validation, and protection against common API security vulnerabilities.
8. Implementing robust logging and monitoring mechanisms to detect and respond to security incidents, including the implementation of Security Information and Event Management (SIEM) Systems.
9. Implementing secure error handling and exception management practices to prevent the leakage of sensitive information and protect against potential security vulnerabilities.
10. Implementing secure communication protocols and encryption mechanisms for secure data exchange between different components and systems.
11. Conducting regular security architecture reviews and risk assessments to identify potential security weaknesses and ensure the effectiveness of implemented security controls.

3.4.4.3 Security Testing and Risk Management

The Contractor duties shall include, but are not limited to:

1. Conducting regular vulnerability assessments and penetration testing to identify and address potential security weaknesses and vulnerabilities in the software systems.
2. Implementing automated security testing tools and techniques to enhance the efficiency and effectiveness of security testing processes.
3. Providing metrics and dashboards for communicating and monitoring threat trend recommendations and status.
4. Providing security expertise in designing, developing, and testing software developed applications, so they are free of security vulnerabilities.
5. Providing guidance on selecting and implementing appropriate security controls and countermeasures to mitigate identified threats.
6. Reproducing, triaging, and addressing security software and application vulnerabilities.
7. Implementing automated code inspection processes to identify defects, common security vulnerabilities, and code standard violations. This includes but is not limited to: Static Application Security Testing, Dynamic Application Security Testing.
8. Analyzing software and application vulnerability findings and identifying the associated risk to the USCB computing environments.
9. Collaborating with stakeholders to prioritize and categorize identified vulnerabilities based on their severity, potential impact, and exploitability.

10. Conducting risk assessments to evaluate the potential consequences of identified vulnerabilities and providing recommendations for risk mitigation and remediation.
11. Monitoring and tracking the progress of vulnerability remediation efforts, ensuring that vulnerabilities are addressed within agreed-upon timelines and according to industry best practices.
12. Conducting post-remediation validation and verification to ensure that vulnerabilities have been effectively addressed and that the software or application is now secure.
13. Documenting and reporting the remediation process for security findings including the identification, assessment, and resolution of vulnerabilities, providing clear and concise reports to stakeholders.
14. Providing expert consultation to development teams in the areas of application security including threat modeling.
15. Collaborating with development teams to identify and prioritize potential threats and vulnerabilities specific to the application being developed.
16. Conducting threat modeling exercises to analyze the application's architecture and identifying potential attack vectors and security weaknesses.
17. Creating the threat models and documentation that capture identified threats, potential impacts, and written recommendations for mitigation strategies.
18. Continuously updating threat models to account for changes in the application's architecture, technology stack, and threat landscape.
19. Conducting regular reviews and assessments of threat models to verify their accuracy and effectiveness in addressing emerging threats.
20. Collaborating with development teams to incorporate threat modeling as an integral part of the software development lifecycle, including design, development, testing, and deployment phases.
21. Conducting regular vulnerability assessments and internal penetration testing to identify and address potential security weaknesses and vulnerabilities in the application.
22. Developing code that is resistant to common vulnerabilities such as injection attacks, cross-site scripting, and cross-site forgery.
23. Validating the implementation of secure:
 - a) Error handling and logging mechanisms are in place to prevent leakage of sensitive information and assist in the identification and resolution of security incidents.
 - b) File handling practices, including proper validation and sanitization of file uploads, secure storage, and protection against path traversal and file inclusion vulnerabilities.
 - c) Configuration management practices are in place including regular patching and updates, secure default settings, proper access controls for system and application configurations.

- d) Session management mechanisms are in place such as session timeouts, secure cookie handling, and protections against session hijacking and fixation attacks.
 - e) Coding practices and guidelines including input validation, output encoding, and proper handling of user authorization and authentication.
24. Participating and complying with incident response processes in accordance with current cyber security policies and procedures.
25. Complying with USCB Secure Coding Standards.

3.4.4.4 Authorization to Operate (ATO)

The Contractor duties shall include, but are not limited to:

1. Developing documentation requirements and supporting evidence.
2. Confirming data types used including, but not limited to, identification, risk implementation, and data handling.
3. Assisting in the development of Privacy Threshold Analysis and Privacy Impact Assessment.
4. Registering systems in the Cyber Security Assessment and Management tool.
5. Assessing and providing controls applicable for ensuring the application or infrastructure is secure in accordance with NIST guidelines and Federal Security Policies.
6. Providing any requested artifacts and evidence collection.
7. Conducting an OIS Assessment (i.e., reviewing and collecting artifacts, interviewing with technical support teams to show application security control compliance and documenting non-compliant findings).
8. Assisting in the assessment and remediation of findings from Independent Penetration Testing results.
9. Documenting security testing findings such as Plan of Action and Milestones (POA&M) in order to mitigate/remediate findings discovered during standard security testing and penetration testing of application security and infrastructure that is needed to support an application.
10. Assisting in FedRAMP certification, as applicable.

3.4.5 Application Operations and Maintenance Execution

The Contractor shall provide Application Operations and Maintenance Execution services to ensure the smooth and efficient operation, maintenance, and support of Decennial Census Program software and applications.

3.4.5.1 Application Tiered Support

The Contractor duties shall include, but are not limited to:

1. Collaborating with stakeholders to leverage log data and relevant metrics from enterprise monitoring systems to report, alert, and informing on operational statistics, performance, and health indicators.

2. Operating, maintaining, and supporting applications in all environments, in alignment with Census enterprise and Decennial security policies, standards, and procedures.
3. Supporting and performing periodic maintenance including but not limited to vulnerability remediation, patching, defect resolution, upgrades, minor enhancements, and providing reports to the appropriate stakeholders.
4. Developing, documenting, implementing, and maintaining procedures, metrics, application documentation on defects, change management, and technical debt reduction.
5. Leading and supporting incident response calls, rapid response teams, root cause analysis, after-action reporting, and troubleshooting.
6. Operating and maintaining purpose-built custom, COTS, GOTS, open-source, geospatial, low-code/no-code, and SaaS applications.

3.4.5.2 Application Monitoring

The Contractor duties shall include, but are not limited to:

1. Designing, documenting, implementing, and maintaining an integrated application monitoring plan, that is integrated across applications, systems, multi-cloud, and on-prem.
2. Collaborating with application teams to ensure that standardized auditing and logging coding standards are in place, to ensure appropriate data is available to provide required monitoring and alerting.
3. Creating and maintaining monitoring, alerting, and data visualizations to enable, at a minimum, monitoring of costs, performance, and availability.
4. Providing periodic assessments and recommendations to improve, at a minimum, costs, performance, and availability.

3.4.5.3 Application FINOPS Operations and Maintenance (O&M)

The Contractor duties shall include, but are not limited to:

1. Collaborating with enterprise FINOPS team to establish naming, tagging, reporting, and cost-alerting standards.
2. Collaborating with application teams to establish reporting, alerting, cost-tracking analysis, spending patterns, cost drivers, and budget reports.
3. Collaborating with applications teams to provide fiscal year budget forecasts.
4. Performing periodic cost analysis reviews and making recommendations to reduce technical costs, improve technical performance, and right size applications.
5. Providing training to application teams on generating reports and custom alerting.
6. Performing cloud migration cost analysis.

3.4.5.4 Continuous Improvement

The Contractor duties shall include, but are not limited to:

1. Continuously improving applications by developing features and managing releases to applications to meet Decennial Census Program's requirements and functionality.
2. Executing change management for continuous improvement as part of the application operations and maintenance.

3.4.6 Cloud Application Migration

The Contractor shall provide Cloud Application Migration services to support the planning, architecture, and execution of the movement of applications or workloads from on-premises infrastructure to external cloud services, or between different external cloud services.

The Contractor duties shall include, but are not limited to:

1. Creating, documenting, and maintaining a cloud application migration value proposition (defined for both business and IT) including but not limited to:
 - a) Building understanding around different use-cases of cloud adoption across business practices.
 - b) Defining action steps to achieve objectives within program schedules.
 - c) Providing a cost-benefit analysis.
 - d) Conducting information impact analysis by understanding the Decennial Census Program's data and metrics aligned to business capabilities.
2. Performing an assessment of cloud service providers and service offerings including but not limited to:
 - a) Assisting the Decennial Census Program in selection of different cloud services, assessing and evaluating different service providers from the enterprise.
 - b) Developing a formal cloud adoption framework aligned with the Decennial Census Program's enterprise architecture and security guideline.
 - c) Building protections against cloud-specific risks into Service Level Agreements / Service Level Objectives; scrutinizing provider responses and addressing suggested changes relative to organizational risk.
 - d) Assisting in the assessment of viable features in networking, security and identity architectures, and tools to enable efficient and effective transitions to the cloud.
3. Guiding/implementing the successful deployment and optimized workload placement via application portfolio optimization, rationalization, and dispositioning.
 - a) Identifying and segregating workloads for migration (e.g., identifying workloads to be migrated first, sequence for those grouped).
 - b) Fostering productive collaboration across enterprise units to contribute to successful deployment.
 - c) Establishing and documenting cloud management workflow; supporting Decennial Census Program cloud management platforms, tools, and central monitoring process.
 - d) Developing, managing, and updating the cloud application migration implementation plan so it is in continual alignment with the project's value proposition.

4. Providing a robust governance process for the cloud application migration including but not limited to:
 - a) Establishing and documenting ownership, responsibility, and risk acceptance by outlining expectations for design, priority, and operations and maintenance of the cloud.
 - b) Supporting cloud cost management (FinOps).
 - c) Establishing and documenting a security control plane to achieve better visibility, data security, threat protection and compliance.
 - d) Developing a life cycle approach to cloud governance that emphasizes operational control in a shared responsibility model.
 - e) Creating an audit trail to increase visibility and accountability for all application-related changes to the cloud environment.
5. Establishing and documenting a Cloud optimization plan and roadmap for FinOps, DevSecOps, and Zero to Mastery (ZTM) disciplines for cloud application migration.
 - a) Creating shared objectives for cross functional teams; emphasizing flexibility, customer centricity, and operational performance and process adherence.
 - b) Using analytics to drive insights and support cloud cost management.
 - c) Iterating to optimize and evolve cloud footprint through service management, enhanced enterprise, and application management.

3.4.7 Research, Development and Innovation

The Contractor shall provide Research, Development and Innovation services to support activities to assess the technical, operational, cost feasibility, and analyses of alternatives (AoA) of new technologies and approaches to meet functional and non-functional requirements.

3.4.7.1 Research and Development

The Contractor duties shall include, but are not limited to:

1. Conducting or participating in feasibility studies to assess the technical feasibility of the proposed technology.
2. Assisting in conducting the cost feasibility, including cost estimates for the development, implementation, and maintenance of the researched technology.
3. Assessing the operational feasibility of proposed technology, considering its impact on existing processes and resources.
4. In support of feasibility studies, considering legal and regulatory feasibility, such that the proposed system complies with relevant laws and standards.
5. Assessing the compatibility of the proposed technology with existing systems and infrastructure.
6. Evaluating the social and organizational impact of the proposed technology on stakeholders and end-users.
7. Considering alternative solutions and providing a comparative analysis to help decision-makers choose the most feasible option.

8. Establishing, documenting, implementing, and maintaining a standard process for conducting feasibility studies/market research.
9. Planning for the evolution of the application portfolio. Creating future versions or prototypes that model use of emerging technologies. Highlighting the changes required to assist in planning.
10. Documenting and reporting on study findings.

3.4.7.2 Application Solution Prototyping and Research

The Contractor duties shall include, but are not limited to:

1. Designing, developing, and documenting application prototypes and pilots to determine feasibility of technologies, emerging or otherwise, to Decennial Census Programs.
2. Documenting research findings and results and making recommendations.

3.4.7.3 Application / Solution Analyses of Alternatives

The Contractor duties shall include, but are not limited to:

1. Conducting AoA using Decennial's AoA process for existing and emerging technologies including, but not limited to, cloud services, methodologies, and products.
2. Maintaining AoA process for solutions and technologies evaluated.
3. Establishing, documenting, and maintaining a master list and schedule of AoAs to be evaluated and current statuses of AoAs.
4. Providing thought leadership to understand the business and missions to propose new and emerging technologies that meet the mission, increase productivity, or reduce cost.

3.5 Task Area 5 – Advanced Data Science and Data Management

The Contractor shall provide Advanced Data Science and Data Management services to include data management, analytics and reporting, and advanced analytics/data science capabilities to support Decennial with both foundational needs and innovation objectives, leveraging the agency's critical data assets that are created, received, and/or transformed by Decennial for mission purposes. Task Area activities will align with Decennial Census Programs' overarching advanced analytics and data science strategic objectives as well as the USCB's enterprise advanced analytics and data science standards, frameworks, and governance. All data science tools/platforms will be provided as Government Furnished Equipment (GFE).

3.5.1 Advanced Data Science

The Contractor shall provide Advanced Data Science services to support leverage cutting-edge techniques and technologies to extract valuable insights and patterns from large and complex datasets.

The Contractor duties shall include, but are not limited to:

1. Managing all activities to align with current advanced analytics and data science standards as defined by Decennial, the USCB, industry best practices, and regulatory requirements. Establishing and documenting new standards as needed.

2. Making recommendations based on industry best-practices for foundational advanced analytics and data science improvements. Collaborating with Decennial using DataOps and other agile, industry-leading practices to streamline and scale the data delivery process.
3. Supporting Decennial with its advanced analytics and data science strategic objectives. Bringing expertise with regards to merging, linking, and/or matching data throughout its lifecycle to support critical business processes.
4. Supporting Decennial Census Program in transitioning advanced analytics and data science capabilities from pilot to production and maintaining those capabilities in production environment.
5. Remaining current with advances in data science. Exploring or suggesting opportunities to apply leading trends and innovation opportunities involving advanced analytics and data science.
6. Supporting the ability to code, develop, and translate languages across data science products.
7. Providing expert data scientists who can develop advanced algorithms, models, and frameworks that leverage the latest advances in technologies including, but not limited to:
 - a) Machine learning
 - b) Deep learning
 - c) Neural networks
 - d) Natural language processing
 - e) GAI, large language models (LLM)
 - f) Pattern recognition
 - g) Data wrangling
 - h) Algorithmic programming
 - i) Digital twin development
 - j) Predictive modeling
8. Leveraging automation and machine learning as part of the integrated IT solutions to manage data, predict scenarios, and make recommendations. The Contractor is encouraged to leverage AI/ML solutions in other aspects of the PWS such as Test Automation, Code Generation, AI operations (AIOps) (for maintenance), machine learning operations (MLOps), large language models (LLMOps), etc.
9. Developing, documenting, and testing prototypes and proofs-of-concept leveraging advanced data science.
10. Creating, maintaining, and using synthetic data when necessary and as decided by Census stakeholders to support the full lifecycle of advanced data science (e.g., development, training models). This includes but is not limited to:
 - a) Conducting data cleaning, filtering, and transformations to create machine learning-ready datasets.

- b) Curating datasets via manual or other means for proof-of-concept studies and benchmarking.
11. Following AI regulation and ethical principles in accordance with industry best practice (namely the NIST AI Framework) to closely manage AI risk and ensure trustworthiness of resulting models and other advanced data science products that use AI/ML.
 12. Conducting necessary activities to ensure advanced data science and AI products are safe, secure, reliant, explainable, interpretable, privacy-enhanced, fair (with harmful biased managed), valid and reliable, accountable, and transparent.
 13. Developing, documenting, and maintaining data science documentation including, but not limited to:
 - a) Comprehensive analysis plans
 - b) Technical reports
 - c) Presentations and slides on conducted work.
 - d) User manuals and READMEs for developed code or toolkits.
 - e) Manuscript preparation for conference or journal submissions.
 - f) Coding and documentation templates and compilations of data science best practices.
 - g) Business and technical documents, standard operating procedures, business cases, newsletters, functional descriptions, system specifications, guidelines, special reports, training materials, and other project deliverables to meet the order requirements.
 14. Developing descriptive and predictive models for relevant use cases to automate and augment general survey efforts including, but not limited to:
 - a) Implementing, adapting, and developing classical machine learning and deep learning models.
 - b) Developing and applying time series models to longitudinal datasets.
 - c) Adapting and using natural language processing models for conducting predictive tasks on natural text.
 - d) Employing models for adaptive survey design.
 - e) Constructing anomaly detection frameworks.
 - f) Configuring semi-supervised learning models.
 - g) Constructing active learning frameworks.
 - h) Applying reinforcement learning techniques.
 - i) Conducting feature selection and feature engineering.
 - j) Evaluating trained models with multiple performance metrics with appropriate test sets and constructing learning curves.

3.5.2 Data Management

The Contractor shall provide Data Management services to establish efficient and effective processes for handling and leveraging data within the Decennial Census Program (regardless of on-premises, cloud, or hybrid architectures).

The Contractor duties shall include, but are not limited to:

1. Collaborating with application leads, data architects, and subject matter experts to create data and analytics products and capabilities to ensure all transactions involving mission data, para data, and metadata are conducted securely, in line with business requirements and functional specifications, and maintain the integrity and lineage of the data.
2. Adhering to data management, analytics, and/or advanced data science policies and procedures established by applicable branches/agencies of the Federal Government.
3. Supporting and advising Decennial on data management standards, processes/procedures, architecture options, data management structures, current and future state data architecture, including any interim states required to transition to the future state, and trends/best practices.
4. Building, integrating, and maintaining data platforms including, but not limited to, data lakes, lake houses, warehouses, marts, hubs, knowledge graph, etc. that can handle data at volumes, varieties, and velocities of varying magnitudes including, but not limited to:
 - a) Operational data
 - b) Graph data
 - c) Transactional data
 - d) Spatial data
5. Planning, designing, developing, implementing, and maintaining appropriate data structures to support business needs.
6. Executing data management structures to implement the integration of frames data with near real-time processing solution.
7. Assisting in documenting authoritative data sources and making recommendations on potential changes or improvements.
8. Enhancing, updating, maintaining, and/or developing data dictionaries, catalogs, and repositories that align with enterprise data standards.
9. Providing standardization and consistency of data, using available data libraries, guidance, and other documentation.
10. Identifying and escalating areas of need for data standardization.
11. Planning for and assisting in the archival and transfer of data to national archives or pre-determined repositories.
12. Coordinating and collaborating across service providers the leveraging of metadata (e.g., data usage, audits, statistics, and lineage) to inform data management decisions.

13. Supporting integration activities as necessary (e.g., processes and procedures to export, transform, and/or load data).
14. Identifying, designing, and/or coordinating opportunities to improve business outcomes with data-driven solutions considered part of industry best practice for data management and analytics.
15. Linking data management requirements to existing architecture or to scope future state changes and innovation.

3.5.2.1 Data Engineering and Integration

The Contractor duties shall include, but are not limited to:

1. Translating data into usable forms in a timely manner to meet business needs.
2. Evaluating cataloging, reusability, governance, compliance, and operations readiness for all data artifacts prior to solution recommendations.
3. Documenting planned and actual data flows, data exchanges, system interconnections and interfaces.
4. Building, integrating, and maintaining data pipelines.
5. Defining and implementing conceptual, logical, and physical data modeling concepts, techniques, and tools.
6. Coordinating across service providers the implementation of data pipelines, integration patterns, and other required tools and platforms.
7. Supporting technical integration to ensure that IT systems and applications work together seamlessly.
8. Identifying integration issues, points of failure, and risks that could impact the integration of IT systems and applications.
9. Collaborating with infrastructure providers to coordinate data integration needs and alternatives based on system requirements for system scalability, reliability, and availability.
10. Assisting and supporting the creation, integration, and maintenance of Interface Control Documents.
11. Validating the design of all SoS interfaces supporting Decennial systems in accordance with the data dictionaries and data lineage work products.

3.5.2.2 Data Quality

The Contractor duties shall include, but are not limited to:

1. Assisting in the development of a scalable data quality discipline that involves addressing tactical data quality issues, as well as the strategic aspects of aligning vision, resources, and business objectives. Critical components of promoting this discipline include but are not limited to:
 - a) Managing scope.
 - b) Data governance requirements for data quality needs.

- c) Essential processes and practices required for monitoring and managing data quality.
- d) Modern data quality technologies.
- 2. Coordinating across service providers to track and measure data quality issues.
- 3. Assisting with the creation, establishment, and publication of Data Integrity Standards, policies, metrics, and frameworks as needed.
- 4. Supporting the elicitation of requirements gathering for data quality and data integrity needs including (but not limited to) data availability/accessibility, timeliness, relevance (i.e., usability and precision), accuracy (i.e., completeness and validity), and consistency to aid business, analytics, and technical teams for Decennial Census Programs.
- 5. Collaborating with application teams and other stakeholders on the implementation of data quality and data integrity standards, best practices, updated policies, etc.

3.5.2.3 Data Lineage

The Contractor duties shall include, but are not limited to:

- 1. Analyzing/tracing data across its lifecycle through various sources, systems, and/or applications to identify issues, anomalies, and inefficiencies.
- 2. Analyzing, identifying root cause, and assisting in the resolution around data lineage, gaps, and inconsistencies.
- 3. Exploring data lineage capabilities including, but not limited to, data governance platforms, graph analytics, and AI/ML solutions to investigate, track, and remediate data lineage issues.
- 4. Developing data lineage for responses, event statuses, and address data, for end-to-end data integration from frame, collection, processing, and reporting to dissemination.
- 5. Assisting in establishing ongoing data lineage monitoring, which may include utilizing available platforms and support in setting up new capabilities.
- 6. Creating, documenting, maintaining, and executing data currency policies, processes, and procedures to ensure timeliness and recency of data sharing, integration, and other processes are aligned to business needs.
- 7. Serving as a liaison and facilitating communications between business, IT, and data teams to collect/understand data lineage requirements.
- 8. Linking data lineage requirements to existing architecture or use it to scope future state changes and innovations.

3.5.2.4 Data Storage and Retention

The Contractor duties shall include, but are not limited to:

- 1. Designing, developing, and managing data management disaster recovery plans and data archiving plans.
- 2. Preserving system and application data in accordance with records management regulations and Decennial Census Program policies.
- 3. Assist in the development and execution of effective data disposition by:

- a) Engaging stakeholders
 - b) Clarifying roles and responsibilities
 - c) Leveraging automation, data stewards and privacy champions
4. Designing, developing, and supporting applications, systems, databases, data stores, and repositories used to access, manipulate, manage, store, and backup data required to support all Decennial operations (e.g., data collection, data tabulation, and data dissemination). This includes but is not limited to:
 - a) Collaborating with infrastructure and applications teams on application and database or data store patching and maintenance strategy, implementation and monitoring and troubleshooting.
 - b) Supporting software development services to include designing, developing, implementing, and maintaining software and databases necessary to access, process, manipulate, manage, and store data used to support the Decennial Census Program programs.
 - c) Installing, configuring, administering, and managing databases and data stores.
5. Working collaboratively with architects, engineers, and applications teams to identify, evaluate, create, manage, and use data storage solutions to maximize value for applications.
6. Assisting in establishing ongoing data storage monitoring to optimize data access, retrieval, and performance.

3.5.2.5 Data Policy, Procedures and Governance

The Contractor duties shall include, but are not limited to:

1. Supporting data governance activities including, but not limited to:
 - a) Preparing analysis on key topics
 - b) Briefing outcomes and summaries to necessary stakeholder
 - c) Identifying/escalating issues that require business input; and
 - d) Supporting data governance meetings and working sessions.
2. Establishing, documenting, and maintaining data standards and policies that align to the enterprise data standards and policies.
3. Coordinating across service providers the documentation of data governance needs (e.g., access, sharing, classification, tagging, and quality).
4. Creating, documenting, and maintaining all necessary governance documentation including but not limited to policies, procedures, processes, and data flow documentation.
5. Coordinating and supporting business leaders across service providers in the development and documentation of processes to describe key information on data assets.
6. Analyzing downstream impact of changes to source data, data fields, schemas, and scripts, on systems, applications, and/or business processes, including negative impacts and how to mitigate or minimize risk.

7. Utilizing downstream impact analyses to create recommendations on how to address issues in the short-term as well as proactively prevent negative impacts in the future.
8. Investigating methodology and applications for privacy-preserving data practices, including but not limited to:
 - a) Expanding synthetic data generation techniques via simulation studies or deep learning models.
 - b) Incorporating differential privacy into data modeling or analyses output.
 - c) Implementing federated learning frameworks for sensitive datasets across multiple data providers.
 - d) Performing computation via homomorphic encryption or secure multi-party computation (MPC).

3.5.3 Business Intelligence, Reporting, and Analytics

The Contractor shall provide Business Intelligence, Reporting, and Analytics services to establish a robust and data-driven decision-making framework within the Decennial Census Program.

The Contractor duties shall include, but are not limited to:

1. Supporting the development, enhancements, and maintenance of data reporting and data discovery models, applications, and frameworks.
2. Providing data visualization support, models, and dashboards.
3. Serving as a liaison and facilitating communication between business, IT, and data teams to collect/understand data analytics and reporting requirements.
4. Linking data analytics and reporting requirements to existing architecture or use it to scope future state changes/innovations.
5. Providing general analytics support, including but not limited to:
 - a) Developing reproducible reports and analyses such as benchmarking for performance and efficiency.
 - b) Conducting data mining analyses, visualizations, and statistical tests on datasets to address customer questions.
 - c) Presenting results to an interdisciplinary audience with varying technical expertise.

3.5.4 Model Development

The Contractor shall provide Model Development services to include leveraging advanced analytics and machine learning techniques to develop predictive and prescriptive models that drive data-driven decision-making and improve business outcomes within the Decennial Census Program.

The Contractor duties shall include, but are not limited to:

1. Designing, developing, and testing algorithms and models used for the Decennial Census Program and document research findings and results to make recommendations to the Government.

2. Developing, training, explaining, maintaining, and securing models including but not limited to:
 - a) AI Models
 - b) Architecture/Engineering Models
 - c) Application Models
 - d) Test Modeling
3. Developing, documenting, and maintaining business process models, interface data flow documentation (e.g., data exchanges and system interconnections), Integrated Operations diagrams, and functional and technical diagrams.
4. Defining a process for the development and validation of Decennial Census Programs Business Process Models (BPMs) that depict the behavior of people, systems, inputs, and outputs.

4.0 Staffing & Key Personnel

DTAM Order Pricing Matrix (Attachment J-XX), outlines the anticipated labor mix for each year of the call order. The Contractor is responsible for ensuring that all personnel are properly trained and meet the minimum requirements for their assigned Labor Categories (LCATs).

The following Key Personnel are essential to the successful execution of the work under this call order:

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Program Manager	Program Manager - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: PMP Certification	Task Area 1

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Systems Architect	Solution Architect - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: The Open Group Architecture Framework (TOGAF), Cloud Architecture certification (AWS/Azure/Google), CISSP	Task Areas 3 & 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Systems Engineer	Systems Engineer - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: Cloud certification (AWS/Azure/Google), Agile certification (Scrum/SAFe), Certified Systems Engineering Professional (CSEP)	Task Areas 2, 3, & 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Solutions Architect	Solution Architect - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: Certified Solutions Architect (AWS/Azure/Google Cloud Professional), Certified Cloud Security Professional, DevOps Certification	Task Areas 3 & 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Data Scientist	Data Scientist - SME	2	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Mandatory Requirements: PhD	Task Area 5

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Application Developer	Application Developer - SME	2	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: Cloud certification (AWS/Azure/Google)	Task Area 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Application Architect	Solution Architect - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Preferred Requirements: Certified Solutions Architect (AWS/Azure/Google Cloud Professional), Certified Cloud Security Professional, DevOps Certifications	

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Security Engineer	Application Security Engineer - SME	1	Experience: Provides technical/management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Has domain and expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. Years of Experience: 15+ Years Mandatory Requirements: Certified Information Systems Security Professional (CISSP) and Certified Cloud Security Professional (CCSP) Preferred Requirements: Cybersecurity certification/Certified Information Security Manager (CISM)/Certified Information Systems Auditor (CISA))	Task Area 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Test Engineer	Test Engineer - Senior	1	Experience: Possesses and applies a comprehensive knowledge across key tasks and high impact assignments. Plans and leads major technology assignments. Evaluates performance results and recommends major changes affecting short-term project growth and success. Functions as a technical expert across multiple project assignments. May supervise others. Years of Experience: 10-14 years Mandatory Requirements: Automated testing experience Preferred Requirements: Automated Testing Certifications (ITSQB Certified Testing/Certified Agile Testing, Certified Software Test Professional (CSTP))	Task Area 4

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Lead Data Architect	Data Architect - Senior	1	Experience: Possesses and applies comprehensive knowledge across key tasks and high impact assignments. Plans and leads major technology assignments. Evaluates performance results and recommends major changes affecting short-term project growth and success. Functions as a technical expert across multiple project assignments. May supervise others. Years of Experience: 10-14 years Preferred Specialized Certs/Requirements: Big data from AWS/Google/Azure, Cloud database, Data Analytics	Task Areas 4 & 5

In alignment with the Decennial BPA PWS requirements, the Contractor will assign and identify the Key Personnel for the entire duration of the call order's period of performance. The Contractor is required to provide resumes for these personnel to demonstrate their qualifications and suitability for their roles.

Any substitutions or replacements of the Key Personnel must adhere to the requirements outlined in the Decennial BPA PWS. This ensures continuity and the maintenance of quality standards throughout the performance of this call order.

5.0 Technical Instructions

Technical instructions to perform effort under the Task Order PWS may be issued in numerical sequence. TIs may be provided to the Contracting Office with the initial Task Order Requirements Package, or with option exercise requests to describe the effort to be performed. A TI may be modified, cancelled, or superseded any time by issuance of a new TI. TIs may be issued during the course of task order performance to provide technical direction that may more closely reflect new information or changed priorities within the TO PWS.

Each TI will establish the effort to be expended for its performance and may include the number of manhours, travel, direct material, and/or other direct costs. The TI shall identify the PWS

paragraph number(s) under which the TI effort is covered. The TI shall clearly be tied to the applicable TO SLIN (if applicable) under which the effort is to be performed. If the TI is used to identify the price of that individual effort, in no event shall the total number of labor hours and price issued under a series of TIs exceed those labor hours and price set forth in section B of the specific task order for a particular SLIN.

6.0 Period of Performance

The period of performance for this call order is one-year base performance period, with seven one-year option periods.

7.0 Place of Performance

Services shall be performed at the USCB located at 4600 Silver Hill Road, Suitland, MD, 20746. Services may be performed at other USCB facilities and remote locations not directly managed by the USCB only with approval of the COR, and subject to any other approvals as required by USCB policy and described below.

The COR may authorize the performance of services under this BPA at facilities not managed by the USCB provided those services do not involve access to or use of: USCB data protected under Title 13 U.S.C.; Federal Tax Information protected under Title 26 U.S.C.; any system that stores or processes data protected under Title 13 or Title 26; or the USCB's corporate information technology (IT) network. Contractors are, at all times, prohibited from accessing any of these data or assets from non-authorized locations.

The COR may authorize performance of services under this BPA (including services that require electronic access to information protected by Title 13 U.S.C. and if authorized by the Internal Revenue Service information protected by Title 26 U.S.C.) at the contractor's individual place of residence. To facilitate performance of these services, access to USCB IT resources may, subject to Division or Office Chief approval, be granted via a Virtual Private Network (VPN) enabled USCB-issued Laptop or through the Virtual Desktop Infrastructure (VDI).

Except as described above, the performance of any services at locations not managed by the USCB that involve access to or use of: data protected under Title 13 U.S.C.; Federal Tax Information protected under Title 26 U.S.C.; any system that stores or processes data protected by Title 13 or Title 26; or the USCB's corporate IT network may be subject to the following additional approvals:

- Review and approval of the physical security controls at the remote location by the USCB's Office of Security. This review may include an on-site inspection.
- Review and approval of the IT security controls in place at the remote location by the USCB's Office of Information Security. This review may involve the authorization of any systems that will handle and process the USCB's data, or that will connect to the USCB's network.

- Review and approval by the Internal Revenue Service if the services to be performed involve access to or use of Federal Tax Information protected under Title 26 U.S.C.
- Review and approval by the USCB's Policy Coordination Office and/or Data Stewardship Executive Policy Committee.

Contractors are not authorized to work remotely with Federal Tax Information (FTI) protected by Title 26

U.S.C. unless executed by federal government contingency planning due to uncertain events and approval from the Internal Revenue Service (IRS). If allowed to work remotely with FTI, each vendor contract must contain language for safeguarding federal tax information as mentioned in IRS Publication 1075, Exhibit 7 (<https://www.irs.gov/pub/irs-pdf/p1075.pdf>).

Contractors may only perform services for the USCB from approved locations. If authorized to work remotely, contractors must still adhere to all applicable USCB Data Stewardship and IT Security Policies and complete all required training. The COR is responsible for ensuring the required training is completed for every contractor authorized to work remotely, and the contractor's access to USCB IT resources and data assets will be terminated for non-compliance.

8.0 Hours of Operation

Unless otherwise noted, under this call order the Contractor is responsible for providing support, Monday thru Friday during normal operating hours (6:00 am-6:00 pm) except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings (unless contractor staff work remotely). All support personnel will be subject to off-hours or on-call requirements as required for maintenance or incident management.

The COR must approve any work done outside of the core regular work hours (6:31PM to 5:59 AM) in advance.

FEDERAL HOLIDAYS

Listed below are recognized Government holidays.

New Year's Day	Labor Day
Martin Luther King Jr. Birthday	Columbus Day
President's Day	Veteran's Day
Memorial Day	Thanksgiving Day
Independence Day	Christmas Day
Juneteenth National Independence Day	Any other day designated by Federal Statute and/or Executive Order

Unless otherwise authorized by the COR, observance of holidays by Government personnel shall not otherwise be a reason for an extension to the PoP, delivery schedule, or entitlement of payment by the Government to the Contractor. In the event the Contractor's personnel (including subcontractor) work during the holiday, they may be compensated for the work in

accordance with the Contractor's operational/employee/business procedures; however, no form of holiday or other premium compensation will be paid by the Government, either as an additional direct or indirect cost, over the payments authorized in the call order.

When the Government grants excused absence to its employees (example: office closing due to inclement weather), onsite Contractor personnel may be dismissed by the Contractor at its discretion. However, such a dismissal will not be an excuse to change any service or delivery requirements under the call order.

9.0 Government Furnished Resources

The Contractor will be responsible for safeguarding all government equipment, information and property provided for Contractor use. The Contractor is not authorized to use GFP/E/I for any purposes other than official Government business as performed under this call order. At the close of each work period, government facilities, equipment, and materials shall be secured.

Contractors working on Government sites will be provided standard business equipment including all or some of the following: desk, chair, phone, computer and access to office equipment such as printers, copiers, fax, etc.

10.0 Travel

The objective of Travel Management is to ensure efficient and effective travel arrangements for contract personnel. Travel requirements are determined by the Decennial Census Program Manager and/or COR. Reimbursement of local commuting expenses is not allowed for travel inside Washington, DC metropolitan area. Local travel is defined as within 50 miles.

The Contractor shall adhere by the following travel stipulations:

1. All travel shall be billed in accordance with the General Services Administration (GSA) Federal Travel Regulation.
2. Travel shall not be taken without prior, written approval of the COR and Government technical lead using the Travel Authorization form.
3. Verifying and validating all travel receipts and Government approvals are included with travel invoices and delivered within five (5) days from return of travel.

Local travel to meetings, conferences or other official business covered under this call order is not reimbursable as a direct charge. Local travel is defined as within 50 miles.

If travel is required, the Government requires the Contractor to obtain prior written approval, using Travel Authorization Form (TAR) (Attachment J-XX), from the Government Team Lead and designated COR to attend meetings, conferences or conduct other official business covered under this TO. The Contractor shall submit requests for travel, using the Travel Authorization Form (TAR), to the Government Team Lead and COR at least 14 business days in advance of the travel. Authorized travel is reimbursable in accordance with the call order and in alignment with (IAW) the Federal Travel Regulation (FTR) and FAR Part 31.205-46 – Travel Costs. The Contractor shall ensure the amount is consistent with the not-to-exceed (NTE) amount specified for this contract line item (CLIN). Travel shall be in accordance with GSA approved rates – i.e. Per Diem, Lodging. A summary of Travel expenses which includes original or legible copies of receipts as prescribed in the FTR shall be submitted with the monthly invoice for the travel occurred.

The FTR may be located and downloaded from: [Federal travel regulation | GSA](#).

Any burden added to the travel cost will be allowed only as defined in the Contractor's standard accounting practice or disclosure agreement.

11.0 Intellectual Property Rights

The existence of any patent, patent application or other intellectual property right that encumbers any deliverable must be disclosed in writing on the cover letter that accompanies the delivery. If no such disclosures are provided, the data rights provisions in FAR 52.227-14 apply.

12.0 Glossary

The definitions of the terms below are specific to the Decennial Program:

Table 12-1: Decennial Glossary

Glossary Term	Description
Integrated Master Schedule (IMS)	IMS provides the program team with a program execution roadmap of meaningful progress and realistic forecasts against a resource-loaded performance measurement baseline. The primary purpose of any IMS is to help the Program Manager and the Program Team optimize the overall execution strategy of a program, coordinate workflows, and assist in the decision-making processes to mitigate risks and resolve challenges on a day-to-day basis.
Integration and Implementation Plans (IIP)	The framework for integrating Information Technology (IT) solutions that make up the Decennial Program programs. It establishes Operational Delivery (OD) and Readiness Reviews.
Line of Business Architecture Documents	These documents outline the vision, design, and technical details of the system, helping ensure interoperability, scalability, and sustainability across different technologies and platforms. Line of business documents to include but are not limited to: 1. Enterprise Architecture (EA) Blueprint 2. System of Systems (SOS) Architecture Document 3. Solution Architecture Document 4. Business Architecture Document 5. Data Architecture Document 6. Integration Architecture Document 7. Security Architecture Document 8. Technology Architecture Document 9. Infrastructure Architecture Document 10. Operational Architecture Document
Operational Delivery (OD)	The technical, logistical, and in-field work that must be coordinated to be deployed within the same timeframe and often using shared system functionality. The focus is on implementing and testing the operation and solution delivery together. Operational Delivery teams, in collaboration with project teams and stakeholders, will define the scope of each OD.

Glossary Term	Description
Runbooks	A comprehensive set of steps detailing system and user interactions when releasing new software or features into a production environment. The runbook is inclusive of selecting the users to participate, coordination with the stakeholders and solution providers involved, monitoring system performance and providing the venue for feedback based on predefined success criteria. The goal is to mitigate risks and provide for a smooth transition to a wider audience.
Readiness Reviews	Planned, documented, management methodology used to evaluate the preparedness of a set of activities that provides visible, objective, and independent evidence that prerequisites have been satisfied, and administrative and technical procedures have been reviewed.
Service Provider	A team responsible for providing specific services or resources, such as database administration, testing, or IT infrastructure.
Solution Provider	A team within Census that builds systems that makes up the Systems of Systems for Decennial 2030.
Systems Engineering Management Plan (SEMP)	The plan that provides the specifics of the technical effort and describes what technical processes will be used, how the processes will be applied using appropriate activities, how the project will be organized to accomplish the activities, information flow within the organization, decision making structure, and the resources required for accomplishing the activities. The process activities are driven by the critical events during any phase of a life cycle (including operations) that set the objectives and work product outputs of the processes and how the processes are integrated
Test and Evaluation Management Plan (TEMP)	The plan that describes the overall structure, objectives, resources, and schedule of the test and evaluation (T&E) program for a system or program. It covers the life-cycle phases from initiation to post deployment and supports major program decisions. It also identifies the tasks, activities, methodologies, responsibilities, and reporting for test activities. The TEMP is the primary test management document and the focal point of the development, review, and approval process for detailed T&E plans. The TEMP is developed and approved in accordance with the relevant policies and guidelines.